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Survey Report 136

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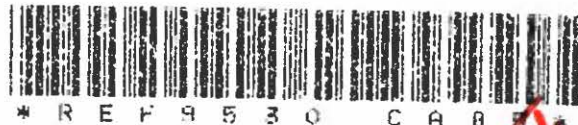
**NEXRAD In-Depth Site Survey for the
Operational Support Facility (OSF)
NORMAN, OKLAHOMA**

Prepared by: DAVID A. LEONE

Prepared for:

DEPARTMENT OF COMMERCE
NEXRAD JSPO, WX7
ROOM 901
8060 13th STREET
SILVER SPRING, MARYLAND 20910

CONTRACT NA-83-SAC-00626



DRAFT

333 Ravenswood Ave. • Menlo Park, CA 94025
(415) 326-6200 • TWX: 910-373-2046 • Telex: 334-486

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By: DAVID A. LEONE

From contributions by:

GEORGIA K. BROWN
WILLIAM A. EDSON
JURIS PETRICEKS

Prepared for:

DEPARTMENT OF COMMERCE
NEXRAD JSPO, WX7
ROOM 901
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SILVER SPRING, MARYLAND 20910

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SRI Project 6273

Approved by:

MURRAY J. BARON, *Deputy Director*
Geoscience and Engineering Center

SRI International

333 Ravenswood Avenue • Menlo Park, California 94025 • U.S.A.
(415) 326-6200 • Cable: SRI INTL MPK • TWX: 910-373-2046

ABSTRACT

This report supplements the NEXRAD Site Survey for the Operational Support Facility (OSF) Norman, Oklahoma, dated July 1984 (SRI Survey Report No. 17). This site is recommended for selection.

The significant recommendations and conclusions of this report are as follows:

- Site Location--Westheimer Airpark.
- Site Configuration--The radar and the OSF office will be connected by a fiber-optic or coaxial cable.
- NEXRAD Tower Geographic Coordinates--Latitude = 35°14'9.75" N; longitude = 97°27'47.25" W.
- Recommended NEXRAD Tower Height (without radome)--20 m.
- Recommended NEXRAD Frequency Assignment--2995 MHz.
- Probability of FAA Approval of NEXRAD Tower--Approved.

A final "Programmatic Environmental Impact Statement" has been published for the NEXRAD Program. To obtain a copy, please see the Preface of this report.

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ACRONYMS

AES	Area Electronic Supervisor
AFB	Air Force Base
A/G	Air/Ground
AGL	Above Ground Level
AM	Area Manager
AP	Anomalous Propagation
ARSR	Air Route Surveillance Radar
ARTCC	Air Route Traffic Control Center
ASDE	Airport Surface Detection Equipment
ASR	Airport Surveillance Radar
ATC	Air Traffic Control
ATCBI	Air Traffic Control Beacon Interrogator
ATCT	Air Traffic Control Tower
AWS	Air Weather Service
BLM	Bureau of Land Management
DF	Direction Finder
DME	Distance Measurement Equipment
DoC	Department of Commerce
DoD	Department of Defense
DoT	Department of Transportation
ECAC	Electromagnetic Compatibility Analysis Center
EMI	Electromagnetic Interference
ET	Electronic Technician
F/O	Fiber Optic
FAA	Federal Aviation Administration
FAAR	Federal Aviation Administration Regulation
HVAC	Heating, Ventilating, Air Conditioning
IFR	Instrument Flight Rules
ILS	Instrument Landing System
INR	Interference-to-Noise Ratio
ISA	Initial Site Assessment
ISS	In-Depth Site Survey
JSPO	Joint System Program Office
LOS	Line of Sight
MIC	Meteorologist-In-Charge
MLOS	Microwave Line of Sight
MLS	Microwave Landing System
MSL	Mean Sea Level
NAS	Naval Air Station
NDB	Nondirectional Beacon
NEC	National Electric Code
NEXRAD	Next Generation Weather Radar
NOAA	National Oceanic and Atmospheric Administration
NOC	Naval Oceanographic Command

ACRONYMS (Concluded)

NTR	NEXRAD Technical Requirements
NWS	National Weather Service
OIC	Official-In-Charge
OSF	Operational Support Facility
PEIS	Programmatic Environmental Impact Statement
PRF	Pulse Repetition Frequency
PSS	Preliminary Site Survey
PUP	Principal User Processor
RACON	Radar Transponder Beacon
RCAG	Remote Communications Air-Ground Facility
RDA	Radar Data Acquisition
RF	Radio Frequency
RFI	Radio Frequency Interference
RPG	Radar Product Generation
RPGOP	Radar Product Generation Operational PUP
RTR	Remote Transmit/Receive Site
TACAN	Tactical Air Navigation
UBC	Uniform Building Code
USFWS	U.S. Fish and Wildlife Service
USFS	U.S. Forest Service
USGS	U.S. Geological Survey
VHF	Very High Frequency
VLA	Very Large Array
VLBA	Very Long Baseline Array
VOR	Very High Frequency Omnidirectional Range
VORTAC	Very High Frequency Omnidirectional Range Tactical Air Navigation
WFO	Warning and Forecast Office

PREFACE

A new nationwide network of weather radars called the Next Generation Weather Radar (NEXRAD) is planned to replace the existing system of weather radars. The mission of the NEXRAD system is to reduce weather-related loss of life, injury, property damage, and interruption of economic activity by the timely acquisition, processing, and distribution of weather radar information. The principal users of the NEXRAD weather information will be the National Weather Service (NWS), the Federal Aviation Administration (FAA), the Air Weather Service (AWS), and the Naval Oceanographic Command (NOC). The first production NEXRAD radars are scheduled to be installed during 1989; installation will continue through the early 1990s.

An Operational Support Facility (OSF) is planned to support the NEXRAD radar network. Initially, the OSF will act as a test bed and a research and development facility to support test and evaluation activities during the acquisition of the prototype NEXRAD radars. The first-article NEXRAD unit will be installed at the OSF site. After Initial Operational Capability (IOC) is achieved, the OSF will provide overall technical and configuration management of the operational NEXRAD system.

Three progressively more-detailed surveys were performed at this site: an Initial Site Assessment (ISA), a Preliminary Site Survey (PSS), and an In-Depth Site Survey (ISS). The ISA was performed by collecting and analyzing maps and other data without visiting the site. During the PSS, the site was visited and a site survey report was published. During the ISS, the site was revisited by an instrumentation van. The instrumentation van was used to take approximately 48 continuous hours of radio-frequency (RF) measurements to aid in selection of a frequency assignment. In addition, a specially-designed video system in the van was used to aid in determining an optimum height for the antenna.

A draft Programmatic Environmental Impact Statement (PEIS) for the NEXRAD program was published in April 1984; public hearings were held in May and the final PEIS was published in November 1984. Copies may be obtained by written request to:

Mr. John Porter
National Weather Service/Code WX7
8060 13th Street
Silver Spring, MD 20910

This In-Depth Site Survey Report expands and updates the Site Survey Report for this candidate site. Although site preparation costs were considered during site selection, detailed site preparation estimates are not included in this report. With the exception of the general site description, pertinent information from the Site Survey Report is not repeated. In the event of conflicting information between the two reports, the information contained in this report should be used because it is more recent.

This ISS Report supports the recommendation for selection contained in the NEXRAD Site Survey Report for the Operational Support Facility (OSF) Norman, Oklahoma, dated July 1984. Changes that have occurred since publication of the Site Survey Report are listed in Appendix A.* Pertinent correspondence and background information contained in Appendix B also update that report. As a convenience, Appendix C contains names, addresses, and telephone numbers of contact points in local organizations and companies. The Electromagnetic Compatibility Analysis Center's (ECAC) supporting analyses of the recommended frequency assignments are contained in Appendix D.

* Information concerning the new NWS/OSF office building has generally not been updated.

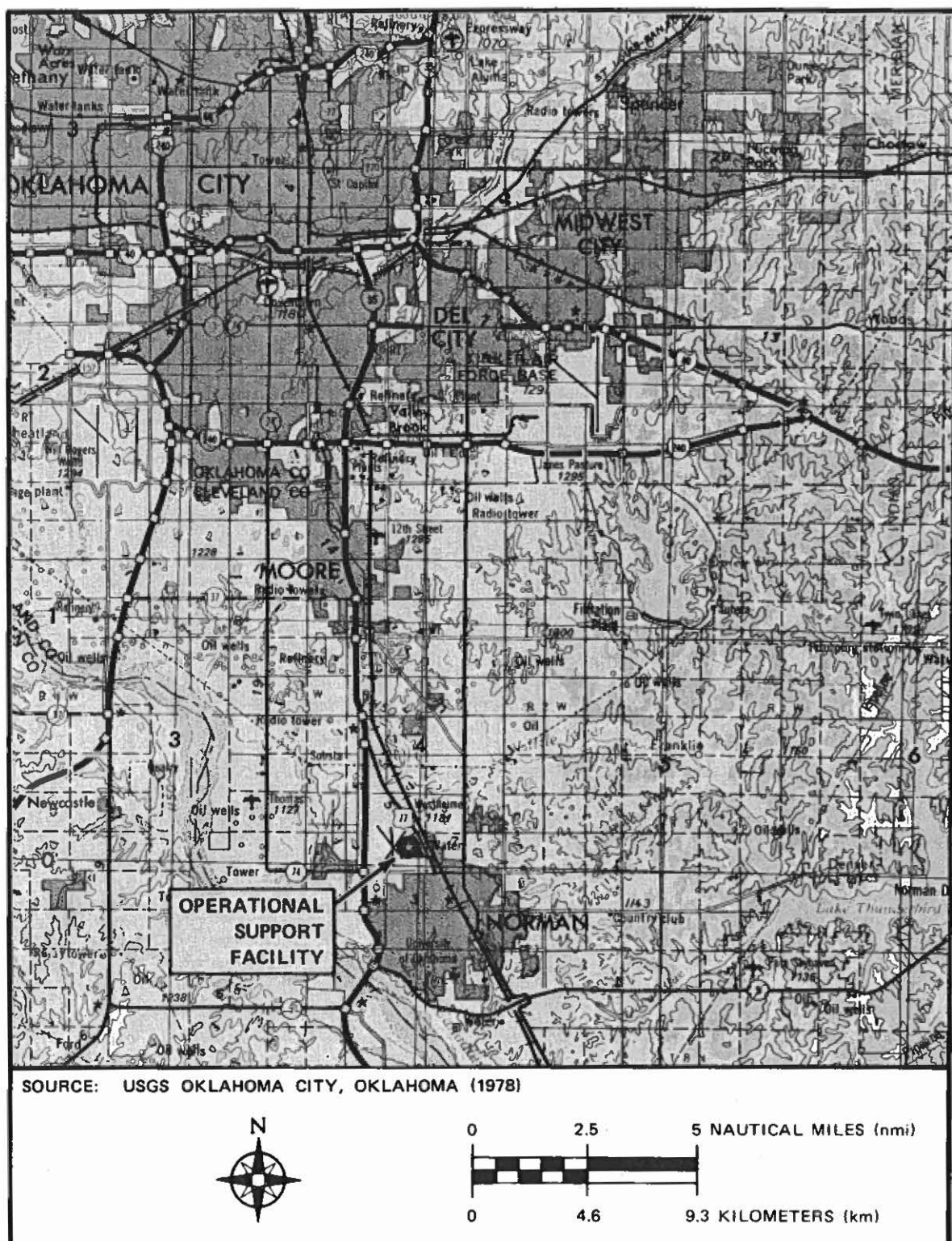
I INTRODUCTION

The proposed NEXRAD site for the Operational Support Facility (OSF) is in Cleveland County, Oklahoma, on the southern portion of Westheimer Airpark (previously called Max Westheimer Airport), about 13 nmi south/southeast of downtown Oklahoma City. Westheimer Airpark is on land owned by the University of Oklahoma.

A new building is currently under construction that will house the OSF office and the Oklahoma City Weather Service Forecast Office (WSFO), which will be relocated from Will Rogers World Airport before the end of 1986. This OSF office will be the NEXRAD Principal User Facility where the OSF NEXRAD display and control equipment will be operated. The OSF NEXRAD would be about 1,000 ft southwest of this new building. A buried fiber-optic or coaxial cable will provide a communications link between the two sites. (The University of Oklahoma has installed a conduit under Halley Drive in anticipation of this requirement.)

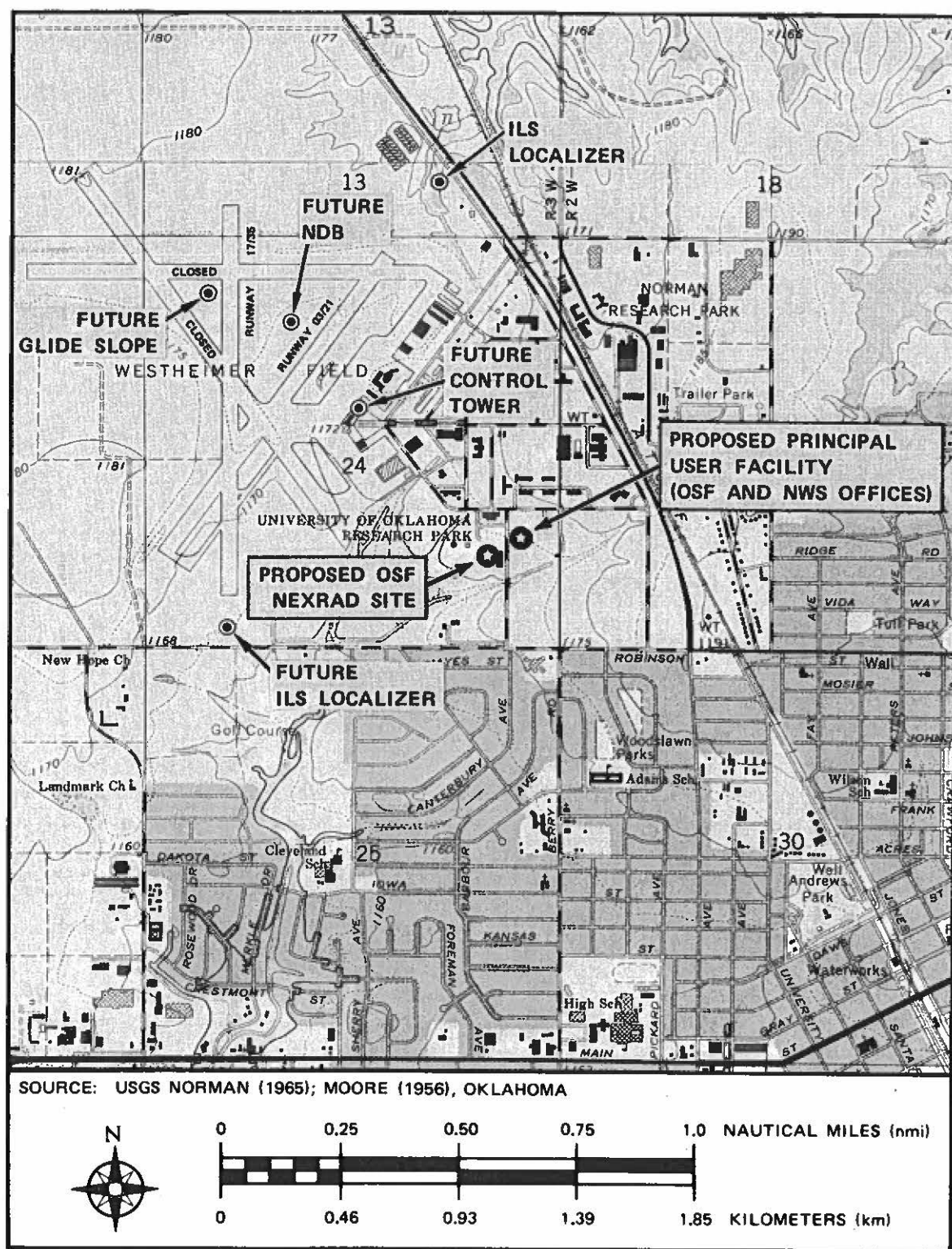
The National Severe Storms Laboratory (NSSL) operates an experimental Doppler radar (hereafter referred to as the Norman Doppler radar) and an experimental WSR-57 radar at Westheimer Airpark. The Norman Doppler radar is approximately 450 ft west-northwest of the proposed OSF NEXRAD site. The experimental WSR-57 is scheduled to be decommissioned before the installation of the OSF NEXRAD.

Location plans, a conceptual site layout, and photographs are shown in Figures 1 through 3. Figure 4 presents a panoramic view of the horizon taken by the NEXRAD van's video system. The video camera was 80 ft above ground level (AGL), which approximates the height to the center of the NEXRAD antenna.



(a) EXISTING AND PROPOSED FACILITIES - 1:250,000 SCALE

FIGURE 1 SITE LOCATION PLAN — PROPOSED OSF SITE AT WESTHEIMER AIRPARK, NORMAN, OKLAHOMA



(b) EXISTING AND PROPOSED FACILITIES - 1:24,000 SCALE

FIGURE 1 SITE LOCATION PLAN — PROPOSED OSF SITE AT WESTHEIMER AIRPARK, NORMAN, OKLAHOMA

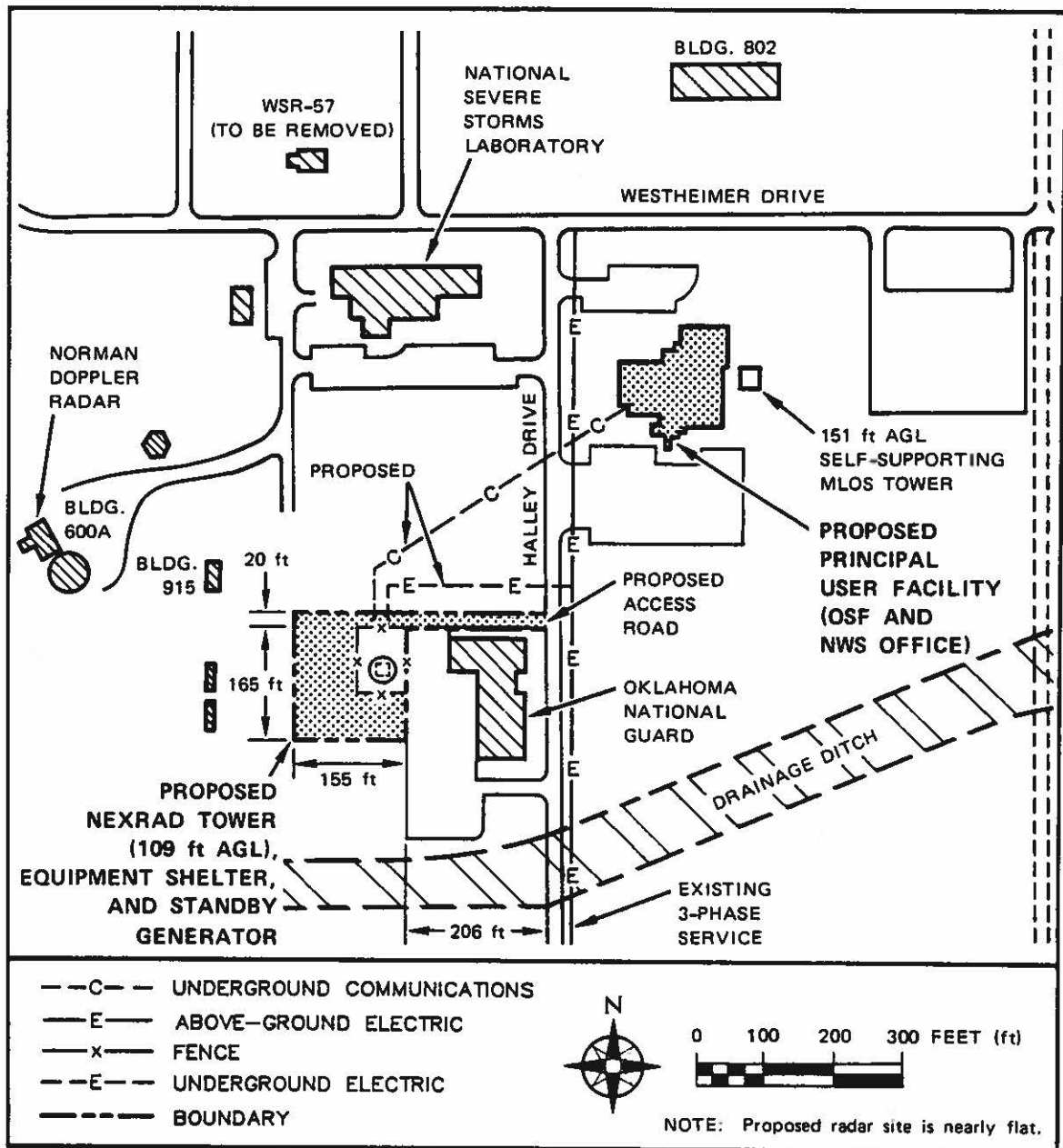
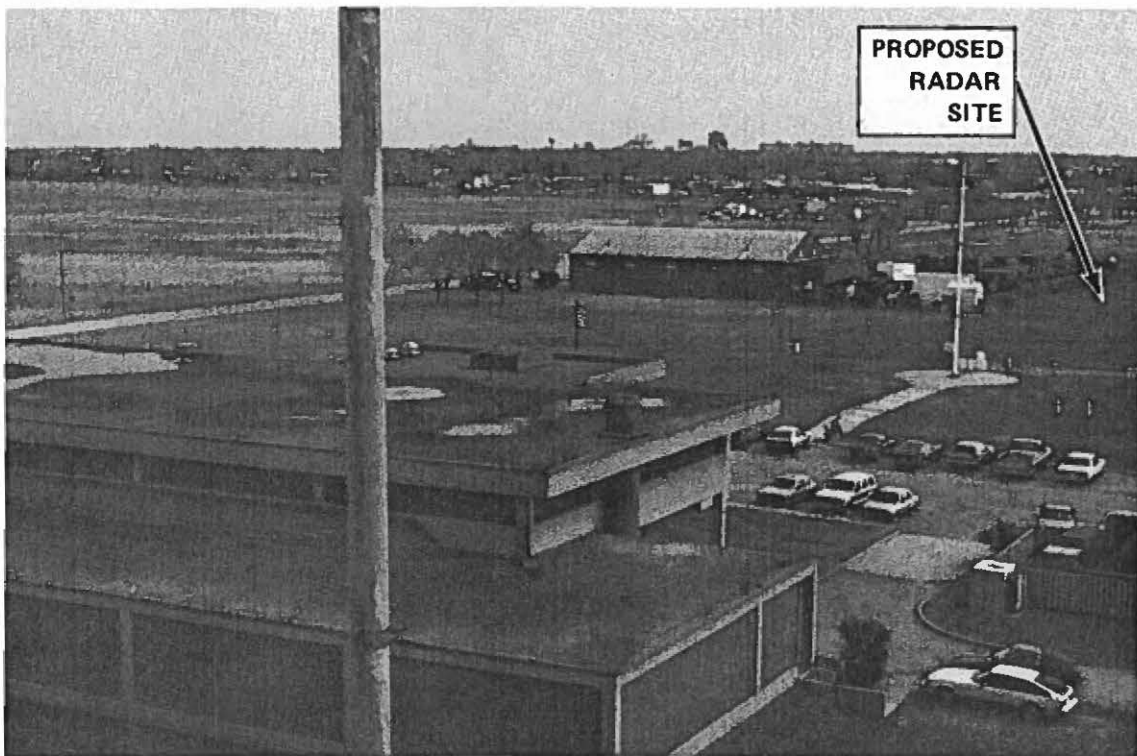


FIGURE 2 CONCEPTUAL SITE LAYOUT AND CONFIGURATION PLAN — PROPOSED OSF SITE AT WESTHEIMER AIRPARK, NORMAN OKLAHOMA

Operational Support Facility
Norman, OK
Rev. 0

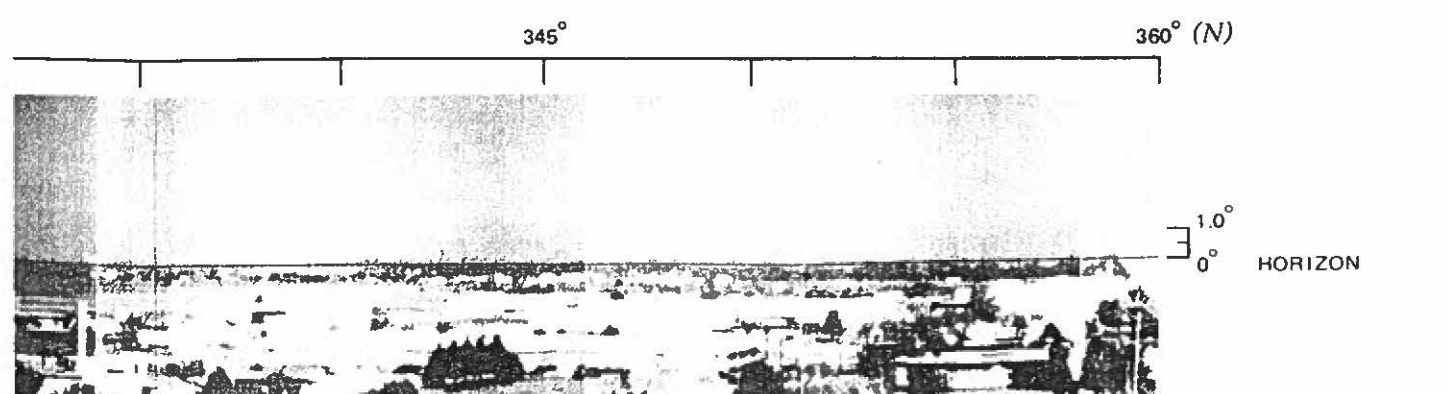
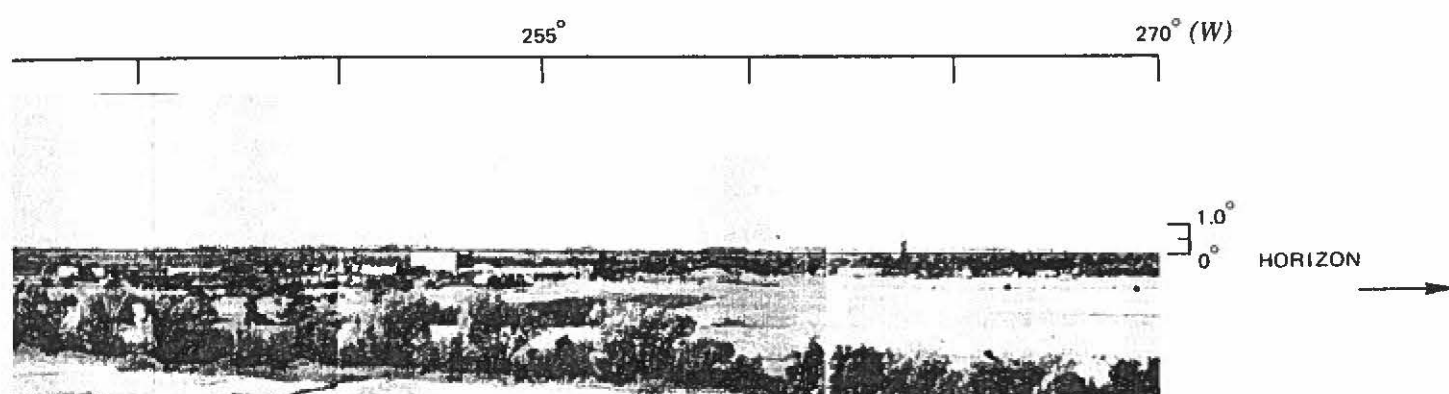
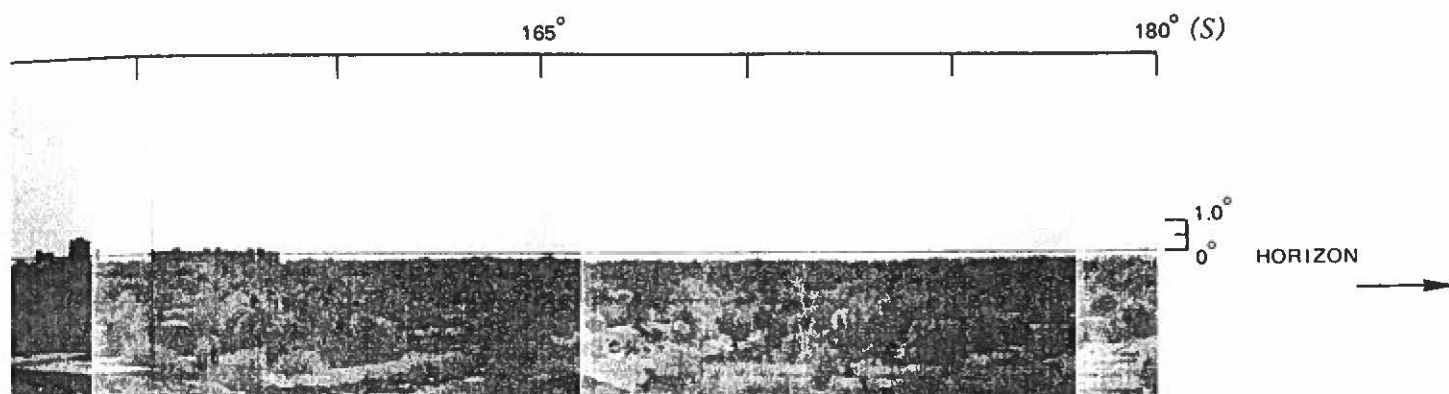
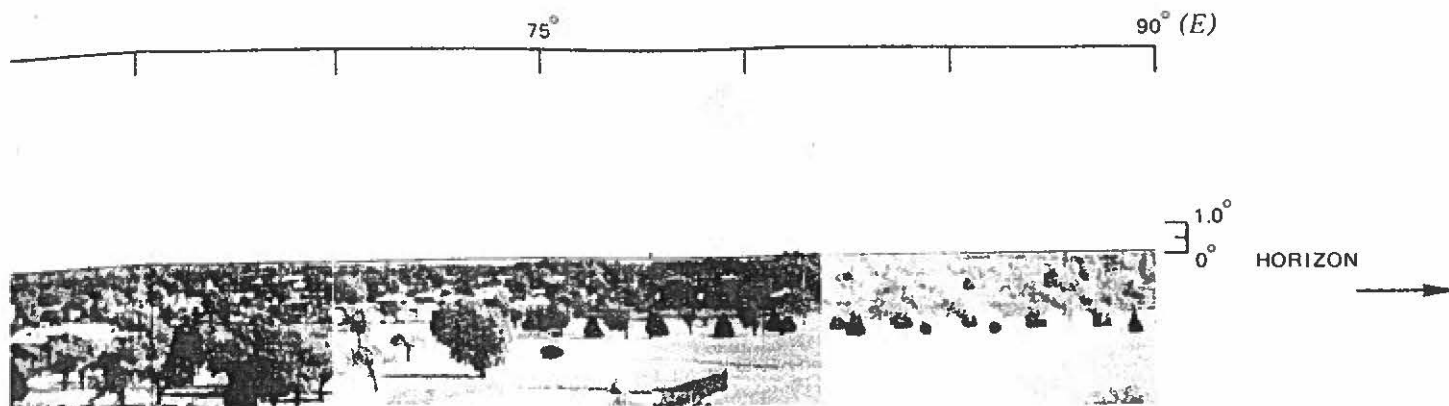


(a) VIEW (LOOKING SOUTHEAST) OF PROPOSED OSF RADAR SITE



(b) VIEW (LOOKING SOUTHEAST) OF NEW NWS/OSF OFFICE BUILDING

FIGURE 3 SITE PHOTOGRAPHS — PROPOSED OSF FACILITIES AT WESTHEIMER AIRPARK, NORMAN, OKLAHOMA



IAN, OKLAHOMA

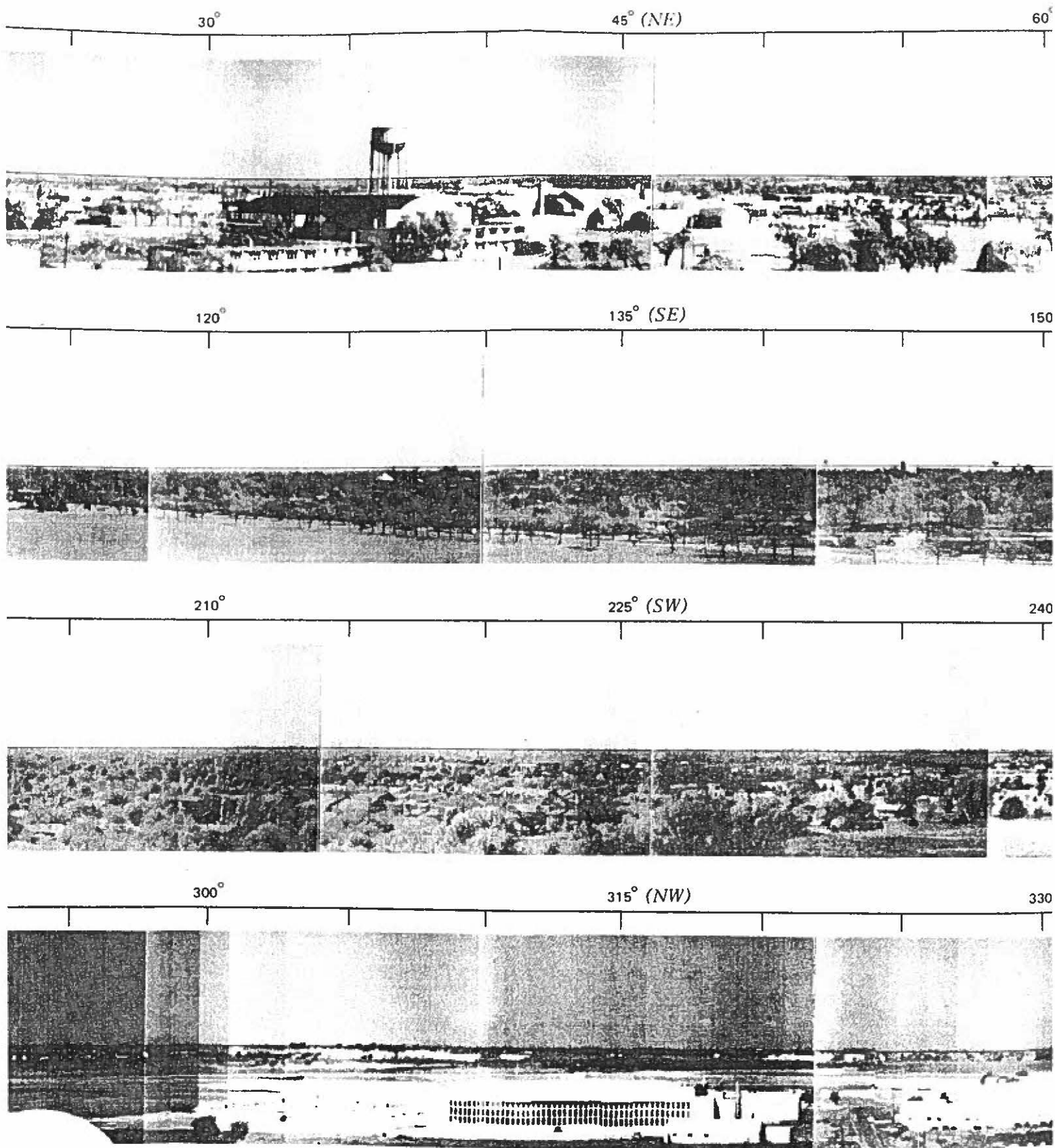
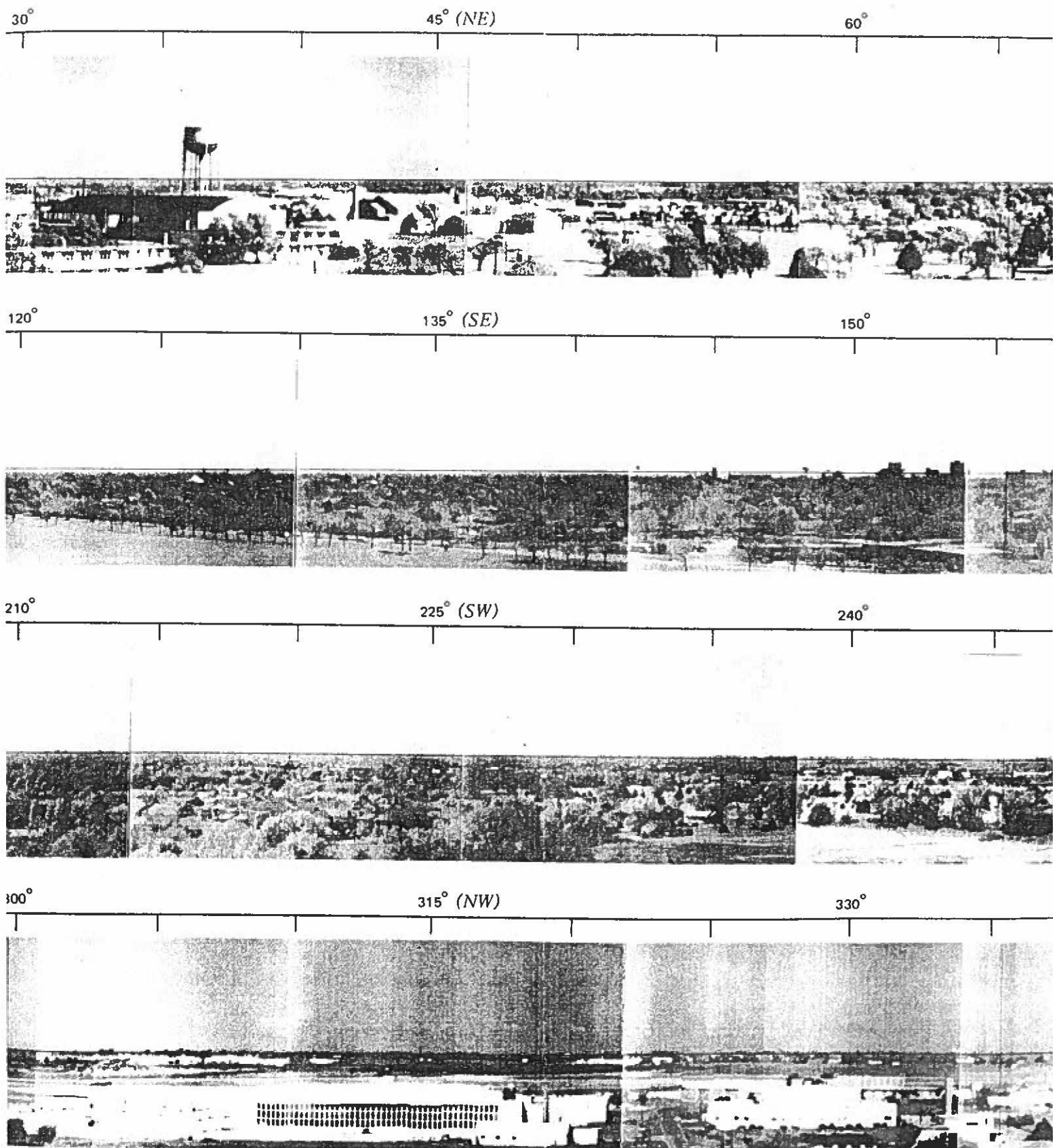


FIGURE 4 PANORAMIC VIEW OF THE HORIZON AT 80 ft AGL — PROPOSED OSF SITE AT WESTHEIMER AIRPARK, I



PANORAMIC VIEW OF THE HORIZON AT 80 ft AGL — PROPOSED OSF SITE AT WESTHEIMER AIRPARK, NORMAN, OKLAHOMA

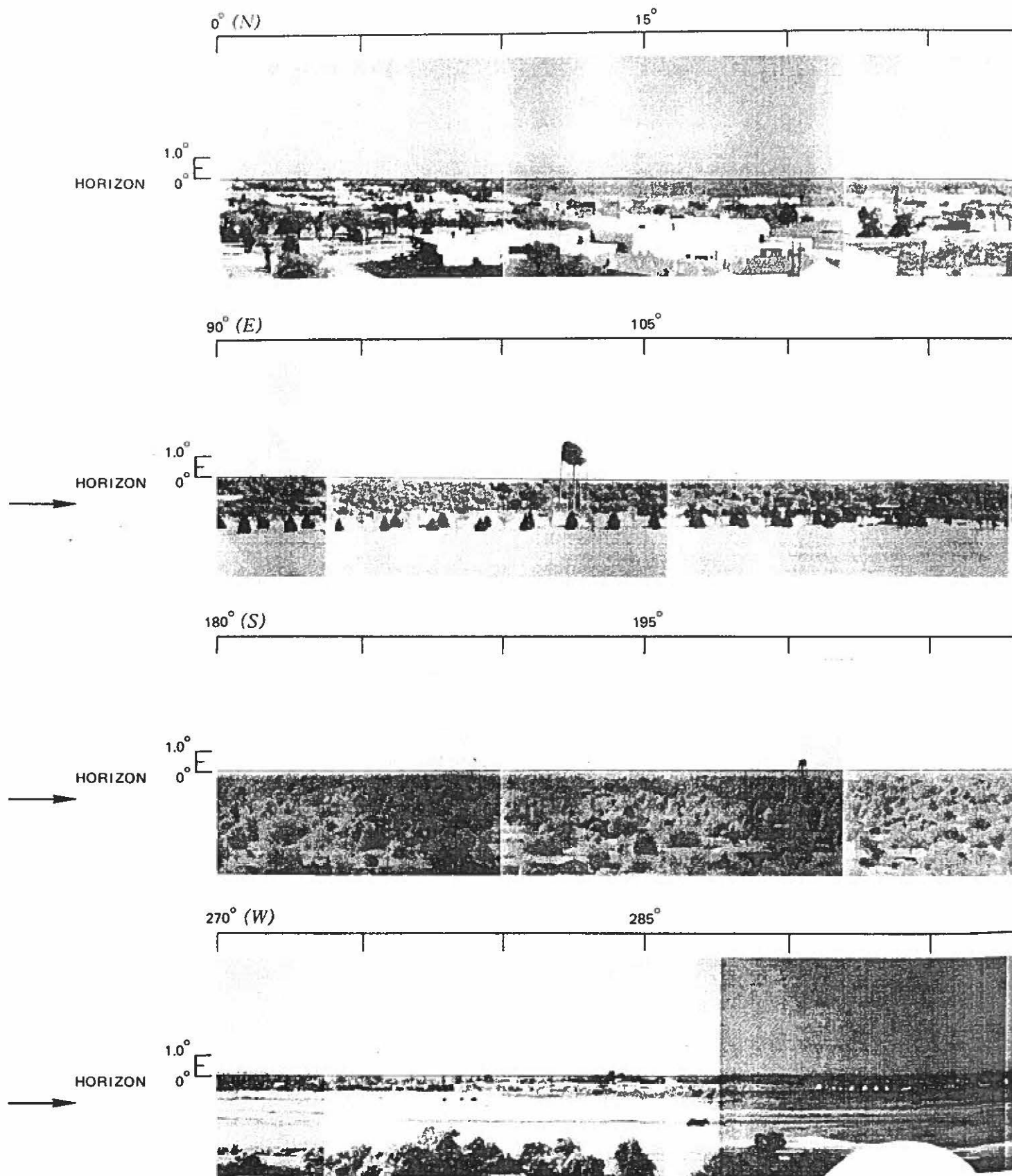


FIGURE 4 PAN

II LONG LEAD-TIME ITEMS

A. Zoning Approval

The University of Oklahoma has granted permission for construction of the OSF NEXRAD at Westheimer Airpark. No zoning approvals are required from the City of Norman for construction on property owned by the University of Oklahoma (Mike Moorman, personal communication, July 1986).

B. Building Permits

Building permits are not required by the City of Norman for construction on University of Oklahoma property unless required by the University. The University will require a building permit for the OSF NEXRAD because it is a commercial installation performed by an outside firm (Mike Moorman, personal communication, October 1986). Plans must be submitted to the Code Enforcement Department, and can be stamped by an engineer registered in a state other than Oklahoma. The City will review the plans only for wind-loading design. The local fire department will review fuel-storage design for compliance with National Fire Protection Association (NFPA) requirements. These reviews normally take two to three days. After the building permit is issued, electrical and HVAC permits will be required. These permits are automatically granted if the electrical and HVAC work is to be performed by a contractor who is licensed and bonded by the City of Norman (Gary Allen, personal communication, October 1986).

C. Power Service at the Radar Site

A lead time of approximately five months would be needed to provide the necessary power service to the proposed site. This length of time is needed for delivery of the transformers (Mike Moorman, personal communication, August 1986).

D. Telephone Service at the NWS Office

Communications lines for NEXRAD were included in the telephone service installed during the construction of the new WSFO/OSF Building. Initially, about 100 of the 250 pairs will be available (Cal Phipps, personal communication, August 1986). About a two-week lead time would be required for the telephone company to activate additional service on an as-required basis (Kathy Henderson, personal communication, October 1986).

III RADAR FREQUENCY ASSIGNMENT

Frequency assignment recommendations are made in the 2.7-to-2.9-GHz band if possible; recommendations are made in the 2.9-to-3.0-GHz band only when analysis shows that all available frequencies in the 2.7-to-2.9-GHz band would cause interference. The OSF NEXRAD frequency assignment recommendations are based on information contained in the Electromagnetic Compatibility Analysis Center's (ECAC) data base and on radio-frequency measurements taken by SRI International at the site. Measurement data were correlated with the ECAC data-base information, and two tentative frequency selections (a primary and a backup) were made--assuming that unseen radiators contained in ECAC's data base were still active but were not operating during the period of SRI's measurements.

As shown in Figure 5, the 2.7-to-2.9-GHz band in the Norman area is extremely congested. The ordinate scale of Figure 5 represents the maximum received pulse power through a 0-dBi-gain antenna. No radiators were seen that were not listed in the ECAC data base. After identifying nearby radiators and analyzing applicable frequency-distance curves,^{*} a 2995-MHz primary frequency assignment is recommended. If this assignment cannot be approved, a secondary assignment of 2970 MHz is recommended. No viable frequency assignments could be identified within the 2.7-to-2.9-GHz band.

ECAC has analyzed the interference-to-noise ratios (INR) between the NEXRAD transmitter and surrounding receivers, and vice versa, for the frequency of 2950 MHz and the backup frequency recommendation of 2970 MHz (see attached report in Appendix D). That report finds both assignments to

^{*} Sangston, Kevin J. and Daniel B. Schwartz, "Technical Characteristics and Frequency-Distance Relationships for NEXRAD and Other Systems in the 2700-3000 MHz Band," Department of Defense Electromagnetic Compatibility Analysis Center, Annapolis, Maryland (February 1985).

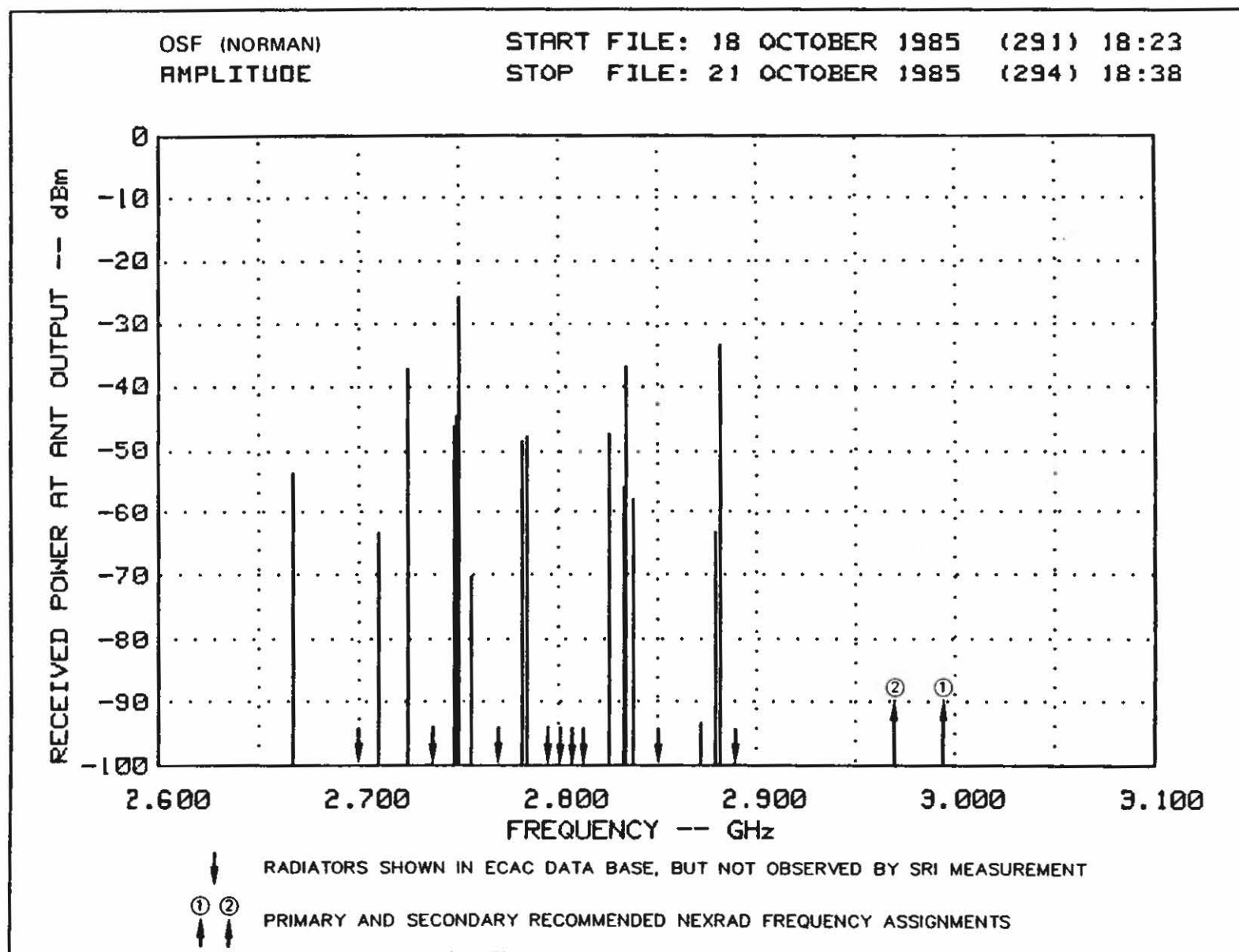


FIGURE 5 FREQUENCY ASSIGNMENT ANALYSIS — PROPOSED OSF NEXRAD AT WESTHEIMER AIRPARK, NORMAN, OKLAHOMA

be viable (Andy Koppenhaver, personal communication, August 1986). ECAC also finds the primary frequency recommendation of 2995 MHz to be viable. Note that the ECAC report in Appendix D recommends 2890 MHz as the primary frequency of the NEXRAD OSF radar at Norman--on the condition that it not operate at the same time as the WDR-73 (Norman Doppler radar). Compliance with this condition was subsequently determined to be too constraining. Because no other frequency assignments could be found in the 2.7-to-2.9-GHz band, both the primary and secondary frequency recommendations for the OSF NEXRAD at Norman are in the 2.9-to-3.0-GHz band.

IV RADAR TOWER HEIGHT

The Site Survey Report recommended a height of 90 ft to the center of the antenna. The corresponding tower height is half-way between two 5-m increment tower-height choices of 20 m and 25 m. The 90-ft height was chosen primarily on the basis of potential EMI with the Norman Doppler. Surrounding terrain and local obstructions would have permitted a lower tower height. The effects caused by the NSSL experimental WSR-57 at Westheimer Airpark are not considered, because this radar will be decommissioned before the OSF NEXRAD is installed.

Assuming a tower height of 20 m is selected, the center of the NEXRAD antenna would be 81 ft AGL, and the bottom of the antenna would be at 67 ft AGL. This provides about a 14.5-ft separation between the bottom of the NEXRAD antenna and the top of the Norman Doppler antenna. At a distance of 450 ft, the NEXRAD beam should be cylindrical; it should start spreading at about 550 ft. Because the OSF NEXRAD is expected to operate at elevations of +0.5 or more deg, the minimum separation between the lower edge of the NEXRAD beam and the top of the Norman Doppler antenna would be 18.5 ft. Thus, the main beam of NEXRAD would not strike the Norman Doppler antenna. Conversely, the main beam of the Norman Doppler will strike the center of the OSF NEXRAD antenna when the Norman Doppler is operating at an elevation of about +5 deg. The worst-case situation occurs when the Norman Doppler antenna is at +5 deg, the NEXRAD antenna is at +0.5 deg, and the two antennas are pointed toward each other. This situation occurs over about 8 deg of azimuth as the two beams sweep by each other. Considering that NEXRAD is at its lowest scan elevation of +0.5 deg only two out of seven scans, and that only one in seven of the Norman Doppler scans is at this most unfavorable elevation of +5 deg, the probability of this worst-case situation is 2×10^{-5} .

Calculations presented in ECAC^{*} and SRI[†] reports indicate that both systems should include highly selective waveguide filters. With this precaution and the recommended frequency separation ($2995 - 2850 = 145$ MHz), there should be no front-end burnout and no serious interference--even for this worst-case situation. Appendix A of the "NEXRAD EMI Study" shows that in the previously described worst-case geometry, NEXRAD would receive spurious signals as large as about 1.5 kW or +32 dBW. The NEXRAD Technical Requirements states that the NEXRAD receiver must comply with requirements contained in the "Manual of Regulations and Procedures for Federal Radio Frequency Management." These requirements specify at least 60 dB discrimination against spurious inputs. Additional discrimination will be provided by the waveguide filter. To meet spurious emission requirements, such a filter will have at least five resonators^{††} and a nominal bandwidth of about 20 MHz. At a frequency of 145 MHz from the passband, such a filter has a calculated attenuation of about 115 dB. For NEXRAD, the minimum detectable signal is about -140 dBW; this value increases by 60 dB (to -80 dBW) for out-of-band signals. Therefore, the calculated signal delivered to the receiver is $+32 - 115 = -83$ dBW, or about 3 dB below the minimum detectable signal. On the basis of this worst-case analysis, it is reasonable to expect that no interference or front-end burnout would be experienced in either system. This expectation is supported by a curve labeled "Typical Diplexer Filter Characteristic--Texas Instruments" that appears in a memorandum (dated 13 February 1979) entitled "ASR-8 Frequency Separation Considerations" that was transmitted to SRI by Churley Jones of JSPO.

^{*} Sangston, Kevin J. and Daniel B. Schwartz, "Technical Characteristics and Frequency-Distance Relationships for NEXRAD and Other Systems in the 2700-3000 MHz Band," Department of Defense Electromagnetic Compatibility Analysis Center, Annapolis, Maryland (February 1985).

[†] Edson, William A., "NEXRAD EMI Study," Special Technical Report 1, Contract NA-83-SAC-00626, SRI International, Menlo Park, CA (April 1984).

^{††} Hinkle, Robert L., "Background Study on Efficient Use of the 2700-3000 MHz Band," National Technical Information Service (NTIS) Report 83-117, U.S. Department of Commerce, Springfield, Virginia (April 1983).

Therefore, a tower height of 20 m (corresponding to a height of 81 ft to the center of the antenna) is recommended.

V FAA REGULATIONS COMPLIANCE

This section of the report presents a partial analysis of the compliance of the proposed NEXRAD facilities with the various Federal Aviation Administration Regulations (FAAR) and requirements. It is intended to provide an indication of the probability of approval by the FAA for construction of the NEXRAD facilities.

JSPO has submitted FAA Form 7460-1, "Notice of Proposed Construction or Alteration," to obtain formal approval. The FAA has reviewed the proposal and has no objection from an airspace utilization standpoint (a copy of the letter of determination is included in Appendix B).

Table 1 presents an analysis of the interference potential between the proposed NEXRAD facilities and the existing and proposed FAA facilities at or near Westheimer Airpark. Most of the areas of potential obstruction listed in this table were derived from FAA Handbook 7400.2C.* This handbook provides general guidance only; the size, height, and radiation pattern of the NEXRAD radar and tower will greatly influence the FAA analyses. Because it is not possible to generate exact criteria, final decisions will be engineering judgements made by FAA personnel. Those cases where more-detailed explanations and analyses are required are presented below.

A. FAAR, Part 77 Requirements[†]

Figure 6 presents an analysis of the proposed NEXRAD tower in relation to the requirements of FAAR, Part 77. The approximate ground elevation

* "Procedures for Handling Airspace Matters," Handbook 7400.2C, Federal Aviation Administration, U.S. Department of Transportation (1 May 1984).

† "Obstruction Requirements," Part 77, Subsection C, "Objects Affecting Navigable Airspace," Federal Aviation Administration Regulations, Department of Transportation (January 1975).

Table 1

ANALYSIS OF INTERFERENCE POTENTIAL FROM NEXRAD FACILITIES TO FAA AND AIRPORT FACILITIES--
PROPOSED NEXRAD OSF TOWER AT WESTHEIMER AIRPARK

FAA Facility	Distance to NEXRAD Facilities	Area of Potential Obstruction	Remarks
Closest Runway	3300'	See FAAR, Part 77	Acceptable-- see Fig. 6
Future ILS Localizer RW 17	95°/3200'	± 35° of runway centerline and 500' from the localizer	Acceptable
ILS Localizer RW 3	49°/4400'	± 35° of runway centerline and 500' from the localizer	Acceptable
Future ILS Glide Slope	Behind GS Ant. > 1000' to CL	Up to 3000' forward of antenna and within 1000' of runway centerline	Acceptable
VORTAC	> 10 nmi	Within 1500' radius and beneath a 1.2° imaginary surface	Acceptable
ASDE	N/A	Within 1500' radius and no shadowing of important ground-control points	No ASDE
ARSR	N/A	Within 3000' radius and no shadowing of ARSR beam	No ARSR
ASR	N/A	Within 3000' radius and no shadowing of important air-control points	No ASR
ATCBI	N/A	Within 3000' radius and no horizontal reflections of beam pattern	No ATCBI
DME	4400'	Within 300' radius	Acceptable
VHF/DF	N/A	Standard type--within 1500' radius Doppler type--within 3000' radius	No DF
NDB	4460'	Within 300' radius	Acceptable
Communications Facilities	> 1500'	Within 1500' radius	Acceptable
Non-Federal Control Tower	> 2000'	View of final approach and surface movement	Unobstructed View

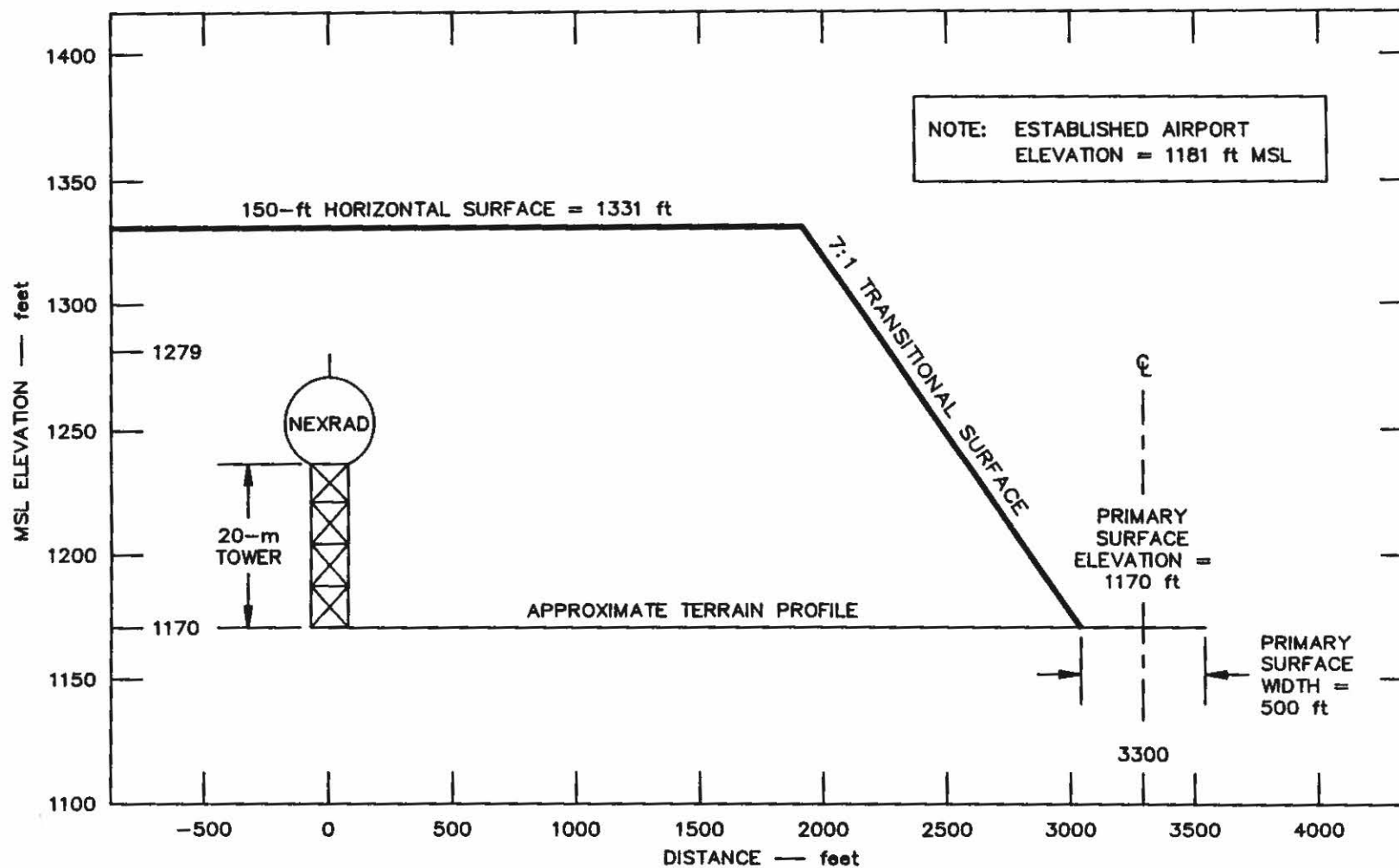


FIGURE 6 FAAR, PART 77 ANALYSIS FOR RUNWAY 03/21 — PROPOSED OSF NEXRAD AT WESTHEIMER AIRPARK TO SERVE THE NORMAN, OKLAHOMA AREA

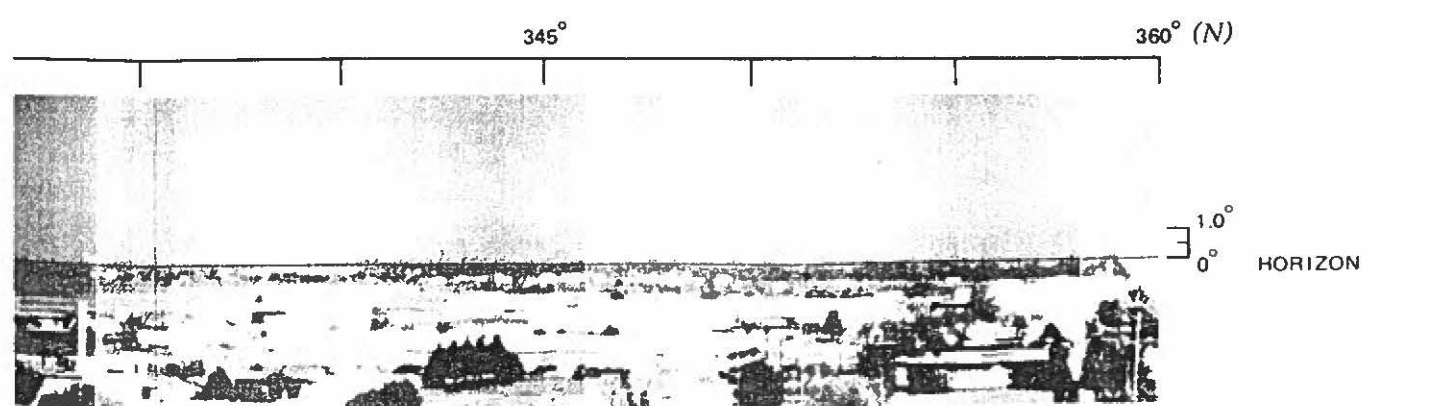
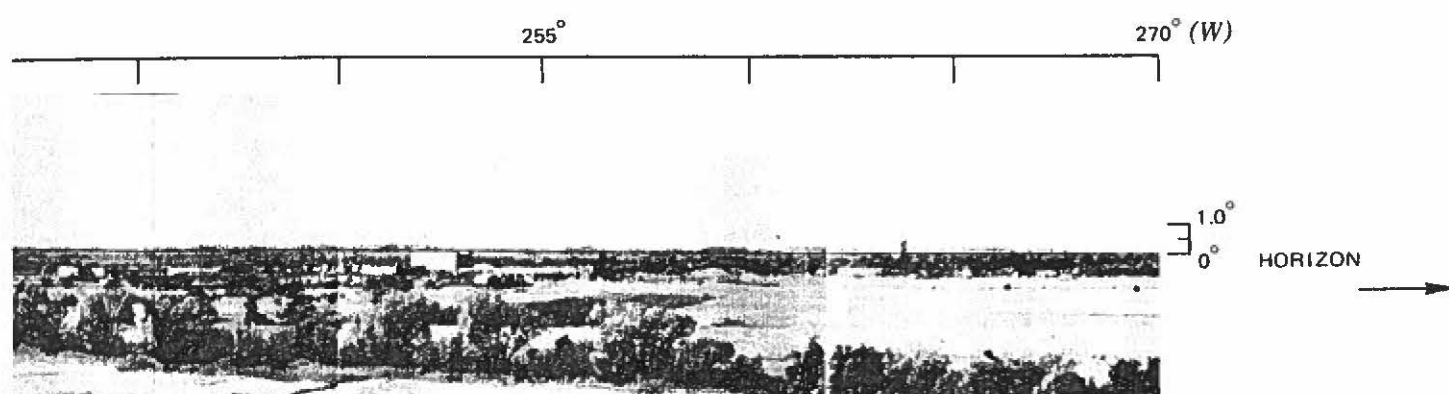
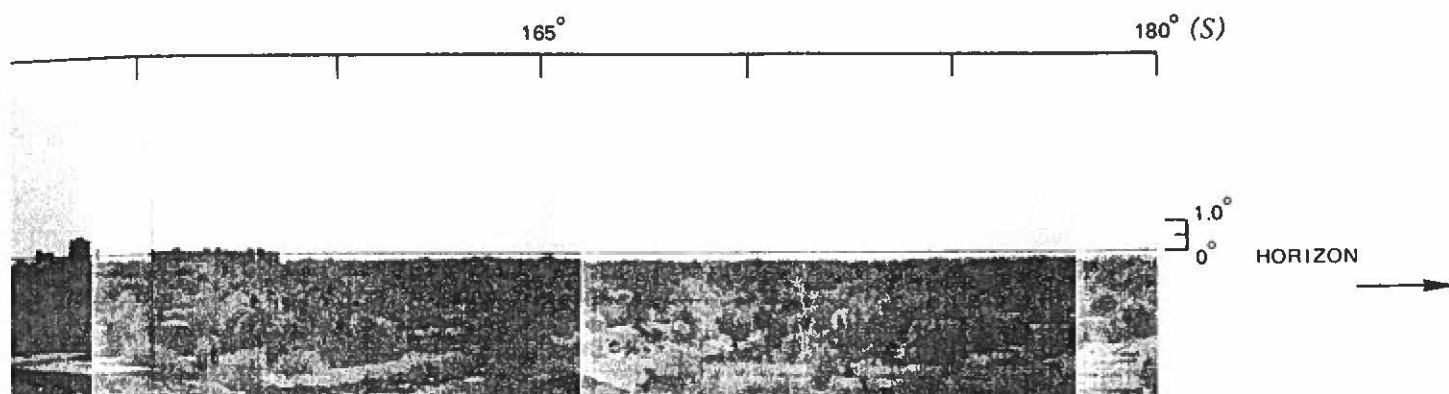
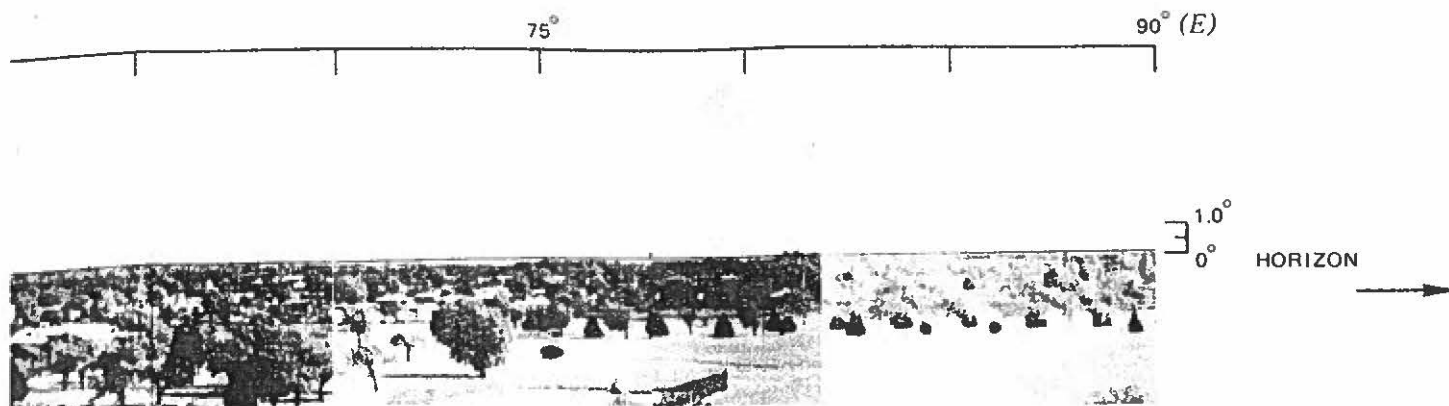
is drawn from the tower perpendicular to the runway. As can be seen, NEXRAD would be within the height limitations established by the Regulations.

B. Obstruction Marking

No marking or lighting is required by FAA Advisory Circular 70/7460-1G for this proposed radar tower. However, the FAA may require lighting and/or marking as a condition of approval after reviewing FAA Form 7460-1.

In summary, no issues that might affect FAA approval have been identified as a result of this analysis.*

* Subsequent to preparation of this report but prior to publication, the FAA approved the 7460-1 application (see Appendix B).



IAN, OKLAHOMA

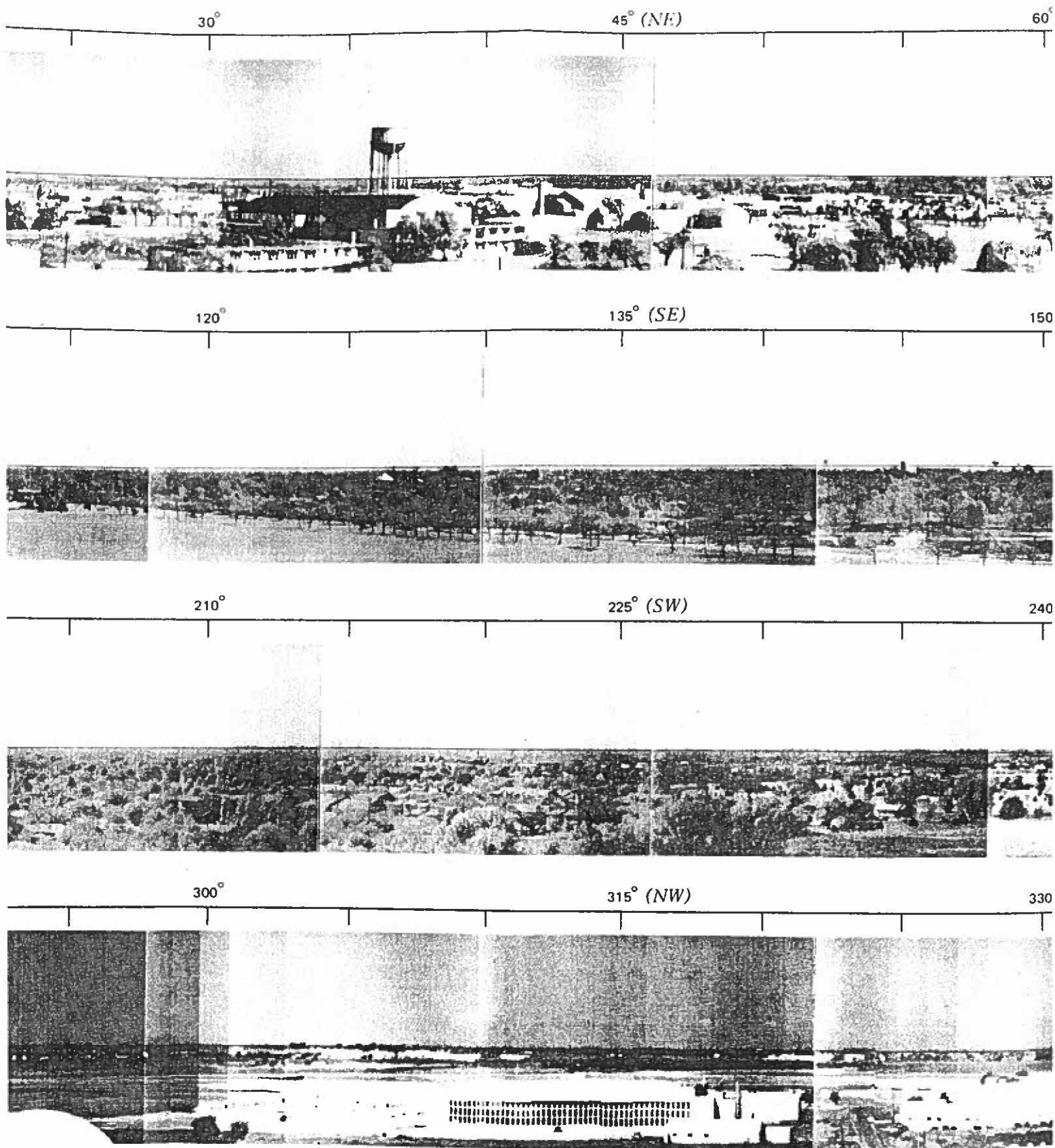
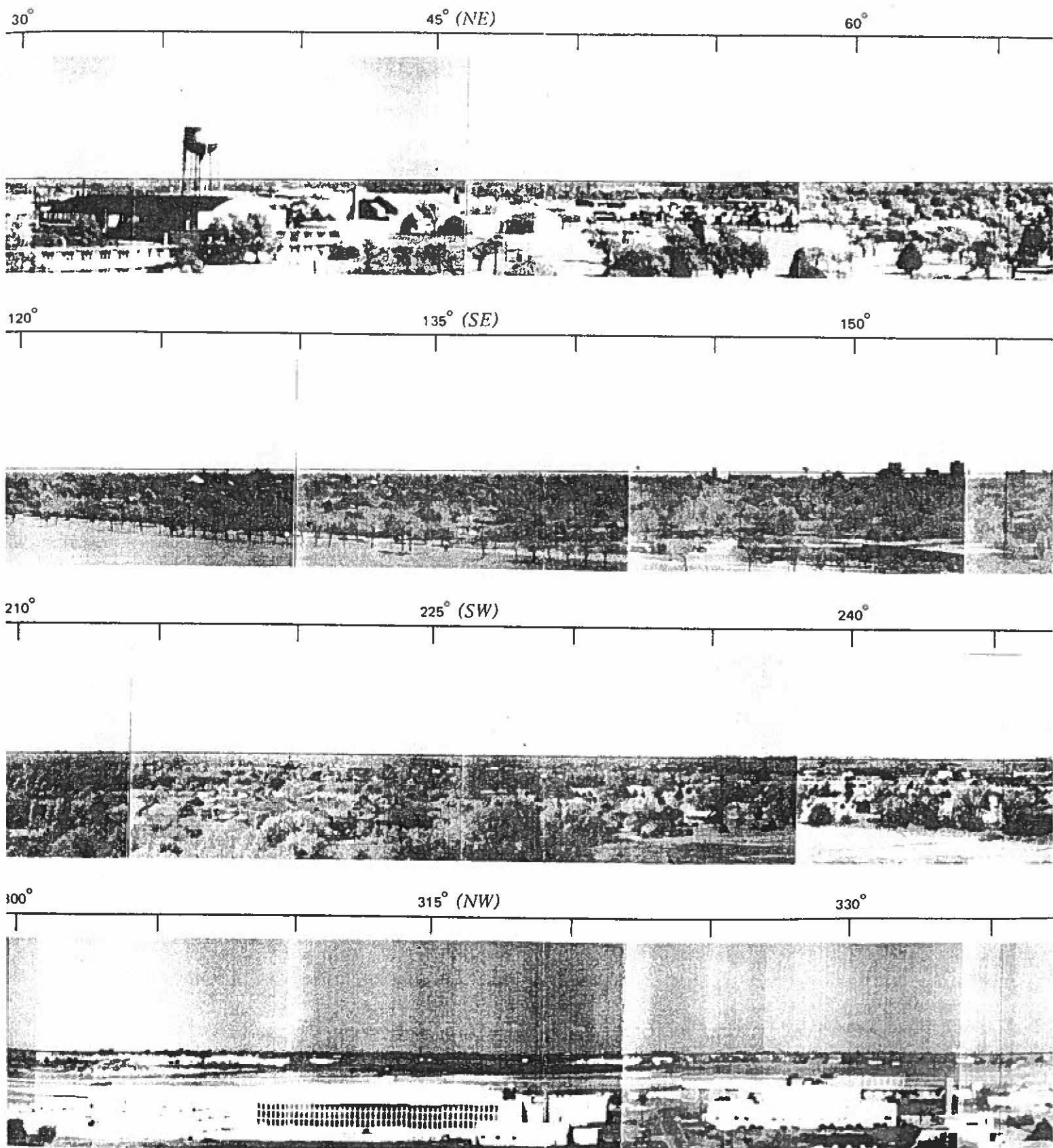


FIGURE 4 PANORAMIC VIEW OF THE HORIZON AT 80 ft AGL — PROPOSED OSF SITE AT WESTHEIMER AIRPARK, I



PANORAMIC VIEW OF THE HORIZON AT 80 ft AGL — PROPOSED OSF SITE AT WESTHEIMER AIRPARK, NORMAN, OKLAHOMA

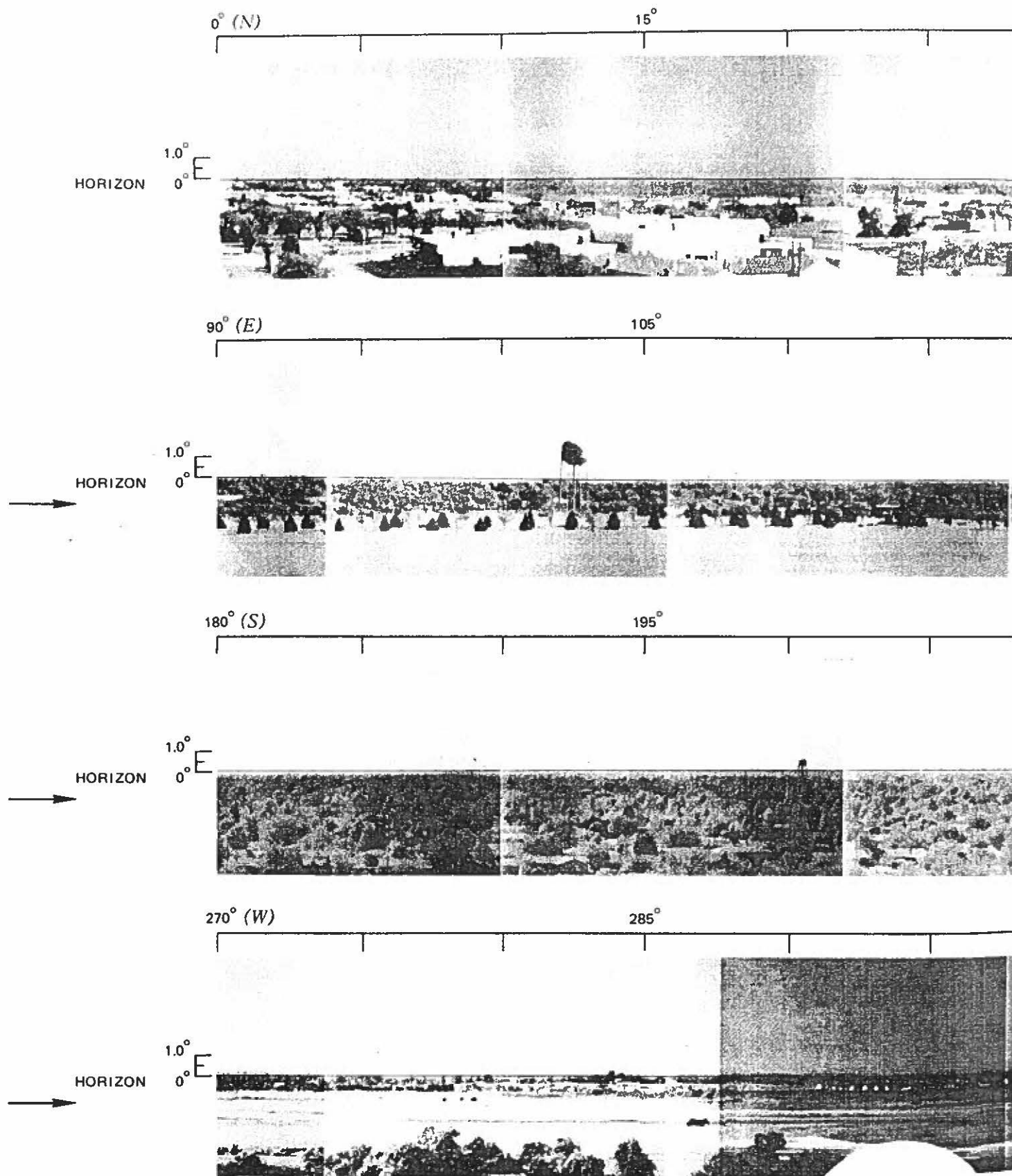


FIGURE 4 PAN

II LONG LEAD-TIME ITEMS

A. Zoning Approval

The University of Oklahoma has granted permission for construction of the OSF NEXRAD at Westheimer Airpark. No zoning approvals are required from the City of Norman for construction on property owned by the University of Oklahoma (Mike Moorman, personal communication, July 1986).

B. Building Permits

Building permits are not required by the City of Norman for construction on University of Oklahoma property unless required by the University. The University will require a building permit for the OSF NEXRAD because it is a commercial installation performed by an outside firm (Mike Moorman, personal communication, October 1986). Plans must be submitted to the Code Enforcement Department, and can be stamped by an engineer registered in a state other than Oklahoma. The City will review the plans only for wind-loading design. The local fire department will review fuel-storage design for compliance with National Fire Protection Association (NFPA) requirements. These reviews normally take two to three days. After the building permit is issued, electrical and HVAC permits will be required. These permits are automatically granted if the electrical and HVAC work is to be performed by a contractor who is licensed and bonded by the City of Norman (Gary Allen, personal communication, October 1986).

C. Power Service at the Radar Site

A lead time of approximately five months would be needed to provide the necessary power service to the proposed site. This length of time is needed for delivery of the transformers (Mike Moorman, personal communication, August 1986).

D. Telephone Service at the NWS Office

Communications lines for NEXRAD were included in the telephone service installed during the construction of the new WSFO/OSF Building. Initially, about 100 of the 250 pairs will be available (Cal Phipps, personal communication, August 1986). About a two-week lead time would be required for the telephone company to activate additional service on an as-required basis (Kathy Henderson, personal communication, October 1986).

III RADAR FREQUENCY ASSIGNMENT

Frequency assignment recommendations are made in the 2.7-to-2.9-GHz band if possible; recommendations are made in the 2.9-to-3.0-GHz band only when analysis shows that all available frequencies in the 2.7-to-2.9-GHz band would cause interference. The OSF NEXRAD frequency assignment recommendations are based on information contained in the Electromagnetic Compatibility Analysis Center's (ECAC) data base and on radio-frequency measurements taken by SRI International at the site. Measurement data were correlated with the ECAC data-base information, and two tentative frequency selections (a primary and a backup) were made--assuming that unseen radiators contained in ECAC's data base were still active but were not operating during the period of SRI's measurements.

As shown in Figure 5, the 2.7-to-2.9-GHz band in the Norman area is extremely congested. The ordinate scale of Figure 5 represents the maximum received pulse power through a 0-dBi-gain antenna. No radiators were seen that were not listed in the ECAC data base. After identifying nearby radiators and analyzing applicable frequency-distance curves,^{*} a 2995-MHz primary frequency assignment is recommended. If this assignment cannot be approved, a secondary assignment of 2970 MHz is recommended. No viable frequency assignments could be identified within the 2.7-to-2.9-GHz band.

ECAC has analyzed the interference-to-noise ratios (INR) between the NEXRAD transmitter and surrounding receivers, and vice versa, for the frequency of 2950 MHz and the backup frequency recommendation of 2970 MHz (see attached report in Appendix D). That report finds both assignments to

^{*} Sangston, Kevin J. and Daniel B. Schwartz, "Technical Characteristics and Frequency-Distance Relationships for NEXRAD and Other Systems in the 2700-3000 MHz Band," Department of Defense Electromagnetic Compatibility Analysis Center, Annapolis, Maryland (February 1985).

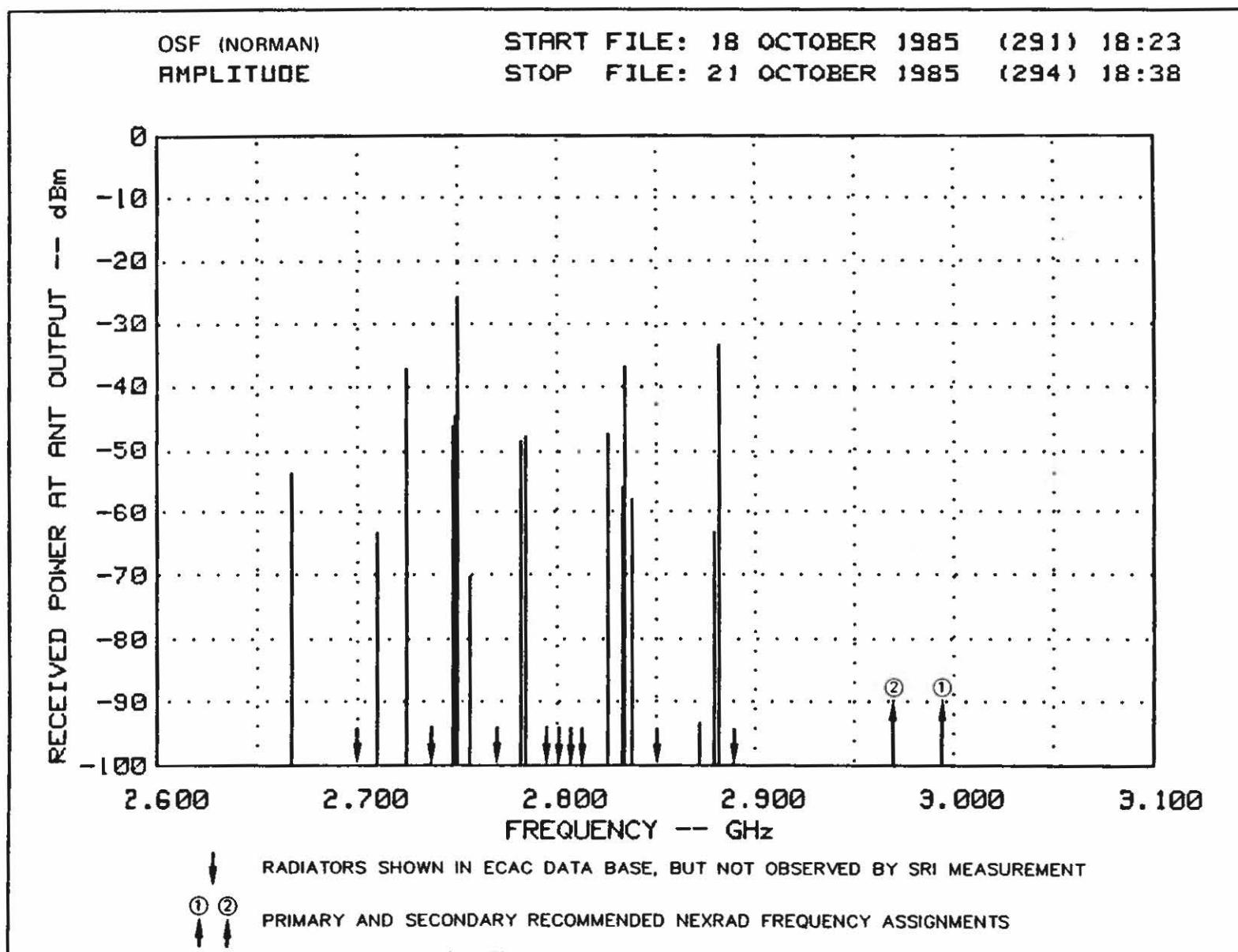


FIGURE 5 FREQUENCY ASSIGNMENT ANALYSIS — PROPOSED OSF NEXRAD AT WESTHEIMER AIRPARK, NORMAN, OKLAHOMA

be viable (Andy Koppenhaver, personal communication, August 1986). ECAC also finds the primary frequency recommendation of 2995 MHz to be viable. Note that the ECAC report in Appendix D recommends 2890 MHz as the primary frequency of the NEXRAD OSF radar at Norman--on the condition that it not operate at the same time as the WDR-73 (Norman Doppler radar). Compliance with this condition was subsequently determined to be too constraining. Because no other frequency assignments could be found in the 2.7-to-2.9-GHz band, both the primary and secondary frequency recommendations for the OSF NEXRAD at Norman are in the 2.9-to-3.0-GHz band.

IV RADAR TOWER HEIGHT

The Site Survey Report recommended a height of 90 ft to the center of the antenna. The corresponding tower height is half-way between two 5-m increment tower-height choices of 20 m and 25 m. The 90-ft height was chosen primarily on the basis of potential EMI with the Norman Doppler. Surrounding terrain and local obstructions would have permitted a lower tower height. The effects caused by the NSSL experimental WSR-57 at Westheimer Airpark are not considered, because this radar will be decommissioned before the OSF NEXRAD is installed.

Assuming a tower height of 20 m is selected, the center of the NEXRAD antenna would be 81 ft AGL, and the bottom of the antenna would be at 67 ft AGL. This provides about a 14.5-ft separation between the bottom of the NEXRAD antenna and the top of the Norman Doppler antenna. At a distance of 450 ft, the NEXRAD beam should be cylindrical; it should start spreading at about 550 ft. Because the OSF NEXRAD is expected to operate at elevations of +0.5 or more deg, the minimum separation between the lower edge of the NEXRAD beam and the top of the Norman Doppler antenna would be 18.5 ft. Thus, the main beam of NEXRAD would not strike the Norman Doppler antenna. Conversely, the main beam of the Norman Doppler will strike the center of the OSF NEXRAD antenna when the Norman Doppler is operating at an elevation of about +5 deg. The worst-case situation occurs when the Norman Doppler antenna is at +5 deg, the NEXRAD antenna is at +0.5 deg, and the two antennas are pointed toward each other. This situation occurs over about 8 deg of azimuth as the two beams sweep by each other. Considering that NEXRAD is at its lowest scan elevation of +0.5 deg only two out of seven scans, and that only one in seven of the Norman Doppler scans is at this most unfavorable elevation of +5 deg, the probability of this worst-case situation is 2×10^{-5} .

Calculations presented in ECAC^{*} and SRI[†] reports indicate that both systems should include highly selective waveguide filters. With this precaution and the recommended frequency separation ($2995 - 2850 = 145$ MHz), there should be no front-end burnout and no serious interference--even for this worst-case situation. Appendix A of the "NEXRAD EMI Study" shows that in the previously described worst-case geometry, NEXRAD would receive spurious signals as large as about 1.5 kW or +32 dBW. The NEXRAD Technical Requirements states that the NEXRAD receiver must comply with requirements contained in the "Manual of Regulations and Procedures for Federal Radio Frequency Management." These requirements specify at least 60 dB discrimination against spurious inputs. Additional discrimination will be provided by the waveguide filter. To meet spurious emission requirements, such a filter will have at least five resonators^{††} and a nominal bandwidth of about 20 MHz. At a frequency of 145 MHz from the passband, such a filter has a calculated attenuation of about 115 dB. For NEXRAD, the minimum detectable signal is about -140 dBW; this value increases by 60 dB (to -80 dBW) for out-of-band signals. Therefore, the calculated signal delivered to the receiver is $+32 - 115 = -83$ dBW, or about 3 dB below the minimum detectable signal. On the basis of this worst-case analysis, it is reasonable to expect that no interference or front-end burnout would be experienced in either system. This expectation is supported by a curve labeled "Typical Diplexer Filter Characteristic--Texas Instruments" that appears in a memorandum (dated 13 February 1979) entitled "ASR-8 Frequency Separation Considerations" that was transmitted to SRI by Churley Jones of JSPO.

^{*} Sangston, Kevin J. and Daniel B. Schwartz, "Technical Characteristics and Frequency-Distance Relationships for NEXRAD and Other Systems in the 2700-3000 MHz Band," Department of Defense Electromagnetic Compatibility Analysis Center, Annapolis, Maryland (February 1985).

[†] Edson, William A., "NEXRAD EMI Study," Special Technical Report 1, Contract NA-83-SAC-00626, SRI International, Menlo Park, CA (April 1984).

^{††} Hinkle, Robert L., "Background Study on Efficient Use of the 2700-3000 MHz Band," National Technical Information Service (NTIS) Report 83-117, U.S. Department of Commerce, Springfield, Virginia (April 1983).

Therefore, a tower height of 20 m (corresponding to a height of 81 ft to the center of the antenna) is recommended.

V FAA REGULATIONS COMPLIANCE

This section of the report presents a partial analysis of the compliance of the proposed NEXRAD facilities with the various Federal Aviation Administration Regulations (FAAR) and requirements. It is intended to provide an indication of the probability of approval by the FAA for construction of the NEXRAD facilities.

JSPO has submitted FAA Form 7460-1, "Notice of Proposed Construction or Alteration," to obtain formal approval. The FAA has reviewed the proposal and has no objection from an airspace utilization standpoint (a copy of the letter of determination is included in Appendix B).

Table 1 presents an analysis of the interference potential between the proposed NEXRAD facilities and the existing and proposed FAA facilities at or near Westheimer Airpark. Most of the areas of potential obstruction listed in this table were derived from FAA Handbook 7400.2C.* This handbook provides general guidance only; the size, height, and radiation pattern of the NEXRAD radar and tower will greatly influence the FAA analyses. Because it is not possible to generate exact criteria, final decisions will be engineering judgements made by FAA personnel. Those cases where more-detailed explanations and analyses are required are presented below.

A. FAAR, Part 77 Requirements[†]

Figure 6 presents an analysis of the proposed NEXRAD tower in relation to the requirements of FAAR, Part 77. The approximate ground elevation

* "Procedures for Handling Airspace Matters," Handbook 7400.2C, Federal Aviation Administration, U.S. Department of Transportation (1 May 1984).

† "Obstruction Requirements," Part 77, Subsection C, "Objects Affecting Navigable Airspace," Federal Aviation Administration Regulations, Department of Transportation (January 1975).

Table 1

ANALYSIS OF INTERFERENCE POTENTIAL FROM NEXRAD FACILITIES TO FAA AND AIRPORT FACILITIES--
PROPOSED NEXRAD OSF TOWER AT WESTHEIMER AIRPARK

FAA Facility	Distance to NEXRAD Facilities	Area of Potential Obstruction	Remarks
Closest Runway	3300'	See FAAR, Part 77	Acceptable-- see Fig. 6
Future ILS Localizer RW 17	95°/3200'	± 35° of runway centerline and 500' from the localizer	Acceptable
ILS Localizer RW 3	49°/4400'	± 35° of runway centerline and 500' from the localizer	Acceptable
Future ILS Glide Slope	Behind GS Ant. > 1000' to CL	Up to 3000' forward of antenna and within 1000' of runway centerline	Acceptable
VORTAC	> 10 nmi	Within 1500' radius and beneath a 1.2° imaginary surface	Acceptable
ASDE	N/A	Within 1500' radius and no shadowing of important ground-control points	No ASDE
ARSR	N/A	Within 3000' radius and no shadowing of ARSR beam	No ARSR
ASR	N/A	Within 3000' radius and no shadowing of important air-control points	No ASR
ATCBI	N/A	Within 3000' radius and no horizontal reflections of beam pattern	No ATCBI
DME	4400'	Within 300' radius	Acceptable
VHF/DF	N/A	Standard type--within 1500' radius Doppler type--within 3000' radius	No DF
NDB	4460'	Within 300' radius	Acceptable
Communications Facilities	> 1500'	Within 1500' radius	Acceptable
Non-Federal Control Tower	> 2000'	View of final approach and surface movement	Unobstructed View

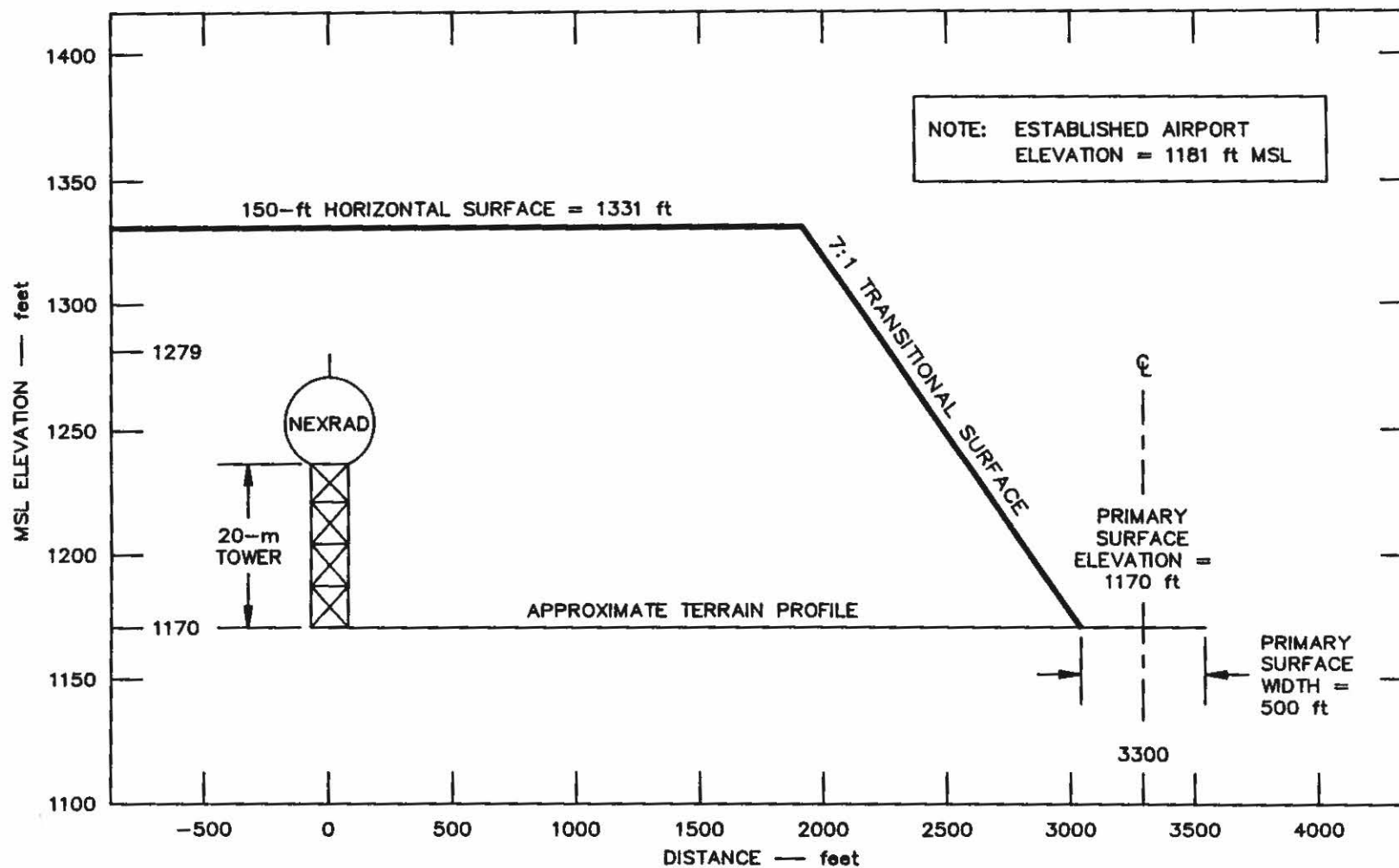


FIGURE 6 FAAR, PART 77 ANALYSIS FOR RUNWAY 03/21 — PROPOSED OSF NEXRAD AT WESTHEIMER AIRPARK TO SERVE THE NORMAN, OKLAHOMA AREA

is drawn from the tower perpendicular to the runway. As can be seen, NEXRAD would be within the height limitations established by the Regulations.

B. Obstruction Marking

No marking or lighting is required by FAA Advisory Circular 70/7460-1G for this proposed radar tower. However, the FAA may require lighting and/or marking as a condition of approval after reviewing FAA Form 7460-1.

In summary, no issues that might affect FAA approval have been identified as a result of this analysis.*

* Subsequent to preparation of this report but prior to publication, the FAA approved the 7460-1 application (see Appendix B).

Operational Support Facility
Norman, OK
Rev. 0

Appendix A

UPDATE TO PRELIMINARY SITE SURVEY REPORT

Appendix A
UPDATE TO SITE SURVEY REPORT

This Appendix contains additional information obtained during the In-Depth Site Survey and updates information that has changed since the time of the Site Survey Report.

General Comments

The airport name "Max Westheimer Airport" has changed to Westheimer Airpark.

The OSF NEXRAD RPG and RPGOP will be in the new NWS/OSF office building at Westheimer Airpark.

Climate

See the 1985 Local Climatological Data, Annual Summary with Comparative Data for Oklahoma City, Oklahoma, in Appendix B of this report (ref. Section III.E of the Site Survey Report).

Geographic Coordinates

The geographic coordinates are: latitude = 35°14'9.75" N; longitude = 97°27'47.25" W (ref. Section III.F).

Future Utility Needs

Since the publication of the Site Survey Report, the University of Oklahoma has upgraded the power along Halley Drive to 12.5 kV 3-phase, which will be adequate to serve NEXRAD and the new NWS/OSF office building. Telephone service will be provided by Southwestern Bell Telephone Company; a 250-pair service to the new NWS/OSF building is presently being installed,

which will be adequate for the OSF NEXRAD. Initially, about 100 of these pairs will be available (ref. Section V.C.2).

Also see Appendix B of this report.

Existing Environmental Conditions

No endangered species are known to exist on the ground at the proposed OSF radar site (Dr. Gary Snell, Director, Oklahoma Biological Survey, personal communication, August 1986). (Ref. Section VII.A.)

Also see Appendix B of this report.

Operational Support Facility
Norman, OK
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Appendix B

CORRESPONDENCE AND BACKGROUND INFORMATION



U.S. Department
of Transportation
Federal Aviation
Administration

OKLAHOMA CITY AIRPORTS DISTRICT OFFICE
204 FAA Building, Wiley Post Airport
Bethany, Oklahoma 73008

MAR 7 1986

Mr. George Hargett
OU Westheimer Airpark
1700 Lexington
Norman, Oklahoma 73069

Dear Mr. Hargett:



OFF	CORR	REF
	✓	OWS
	✓	TWP HZ
	✓	ISA Cel
		WAS
		QWP
		ALL

The Federal Aviation Administration has completed Aeronautical Study 85-ASW-2073-OE for a proposed antenna tower on Westheimer Airpark.

We have reviewed the proposal and have no objection from an airspace utilization standpoint. This was approved with the changed coordinates of 35°14'9.75" latitude and 97°27'47.25" longitude.

Please inform the proponent, SRI International, of our findings.

Sincerely,

Daniel W. Langford

Bill J. Howard
Manager, Oklahoma City
Airports District Office



B-2

Edward Warren: First American Aloft

M E M O R A N D U M

TO: Dave Leone February 24, 1987
FROM: John Chamberlain cc: Teri Stewart
SUBJECT: State of Oklahoma Fish and Game Response to the Proposed
Norman Site

John Skeen, biologist with the State of Oklahoma was briefed by phone on the location and structure height for the Norman site. He states that in view of the developed area in which the site would be located no known habitats will be affected and no potential impacts are to endanger or threatened species is anticipated.

WILDLIFE CONSERVATION COMMISSION

JOHN D. GROENDYKE
CHAIRMAN

K. E. PENNINGTON
VICE-CHAIRMAN

JOHN GIBBS
SECRETARY

ROY BOECHER
MEMBER

DON R. GREENHAW
MEMBER

"H. B." ATKINSON
MEMBER

L. B. SELMAN
MEMBER

CARL PIERCEALL
MEMBER



STEVEN ALAN LEWIS, DIRECTOR

CHARLES R. WALLACE, ASSISTANT DIRECTOR

DEPARTMENT OF WILDLIFE CONSERVATION

1801 N. LINCOLN

P. O. BOX 53465

OKLAHOMA CITY, OK 73152

PH. 5

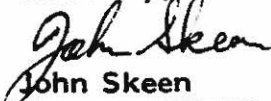
February 27, 1987

John Chamberlain
Research Analyst
SRI International
333 Ravenswood Ave.
Menlo Park, CA. 94025

Dear Mr. Chamberlain:

I reviewed the information you provided on the proposed Norman and Tulsa sites for the NEXRAD facilities. The sites do not contain endangered or threatened species habitat. The only endangered species which is likely to occur near the sites is the bald eagle, which should not be affected significantly by the project.

Sincerely,


John Skeen
Nongame Wildlife Biologist
JS/dl

B-4



**UNITED STATES
DEPARTMENT OF THE INTERIOR
FISH AND WILDLIFE SERVICE**

Ecological Services
222 S. Houston, Suite A
Tulsa, Oklahoma 74127

February 2, 1987

#2-14-87-I-46

Mr. John Chamberlain
SRI International
333 Ravenswood Avenue
Menlo Park, California 94025

Dear Mr. Chamberlain:

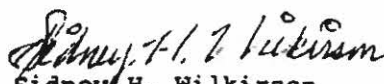
This responds to your letter of January 27, 1987, requesting information on species federally listed or proposed to be listed as threatened or endangered in the vicinity of proposed Next Generation Weather Radar System (NEXRAD) sites in Oklahoma. The sites are in Norman at Westheimer Airpark and 32 km east of Tulsa (northwest of Inola) on the Shreck farm.

The only listed or proposed species which potentially could occur at the NEXRAD sites is the endangered whooping crane (Grus americana). The migration corridor of this crane includes Cleveland County in which the Norman site is located. However, in view of the metropolitan location and lack of suitable stopover habitat, the whooping crane would not be affected by the proposed action.

Construction of towers and radar domes such as proposed pose a danger to various migratory birds during migration flights, especially during bad weather. This potential threat needs to be studied and any appropriate protective measures incorporated into the projects.

If we can be of further assistance, please call our office at 918/581-7458.

Sincerely yours,


Sidney H. Wilkirson
Field Supervisor

cc:
Regional Director, FWS, Albuquerque, New Mexico (AWE)

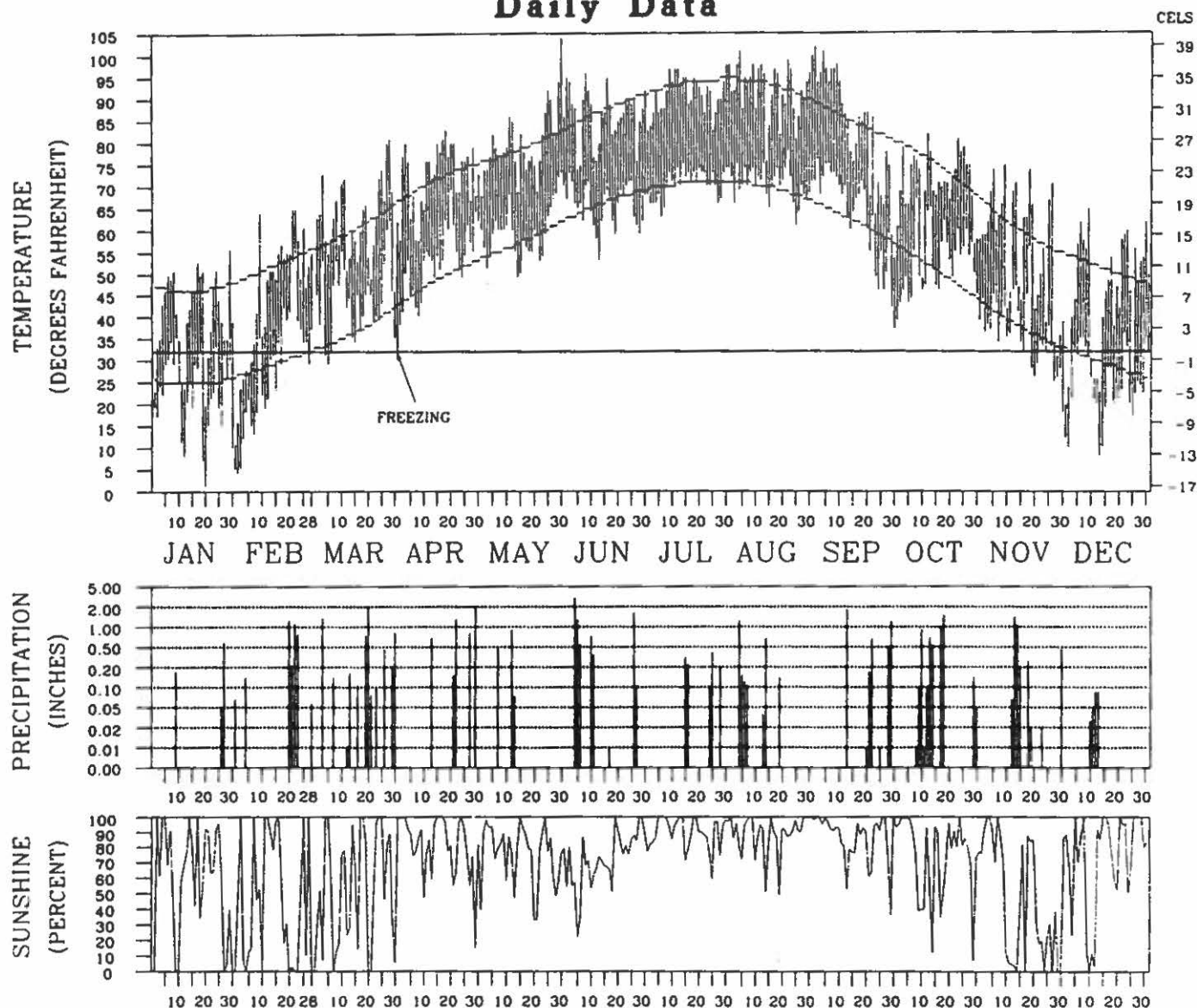


1985 LOCAL CLIMATOLOGICAL DATA

ANNUAL SUMMARY WITH COMPARATIVE DATA

OKLAHOMA CITY, OKLAHOMA

Daily Data



TEMPERATURE DEPICTS NORMAL MAXIMUM, NORMAL MINIMUM AND ACTUAL DAILY HIGH AND LOW VALUES (FAHRENHEIT)
 PRECIPITATION IS MEASURED IN INCHES. SCALE IS NON-LINEAR
 SUNSHINE IS PERCENT OF THE POSSIBLE SUNSHINE

I CERTIFY THAT THIS IS AN OFFICIAL PUBLICATION OF THE NATIONAL OCEANIC AND ATMOSPHERIC ADMINISTRATION, AND IS COMPILED FROM RECORDS ON FILE AT THE NATIONAL CLIMATIC DATA CENTER, ASHEVILLE, NORTH CAROLINA, 28801

noaa

NATIONAL
OCEANIC AND
ATMOSPHERIC ADMINISTRATION

NATIONAL
ENVIRONMENTAL SATELLITE, DATA
AND INFORMATION SERVICE

NATIONAL
CLIMATIC DATA CENTER
ASHEVILLE NORTH CAROLINA

Kenneth D. Halpern
 DIRECTOR
 NATIONAL CLIMATIC DATA CENTER

METEOROLOGICAL DATA FOR 1985

OKLAHOMA CITY, OKLAHOMA

LATITUDE: 35°24' N LONGITUDE: 97°36' W ELEVATION: FT. GRND 1285 BARO 01283 TIME ZONE: CENTRAL WBAN: 13967

	JAN	FEB	MAR	APR	MAY	JUNE	JULY	AUG	SEP	OCT	NOV	DEC	YEAR
TEMPERATURE °F:													
Averages													
-Daily Maximum	39.6	46.1	62.6	72.7	80.2	86.5	91.4	92.3	83.3	70.5	56.1	45.9	68.9
-Daily Minimum	21.6	28.2	43.3	52.7	59.7	65.4	70.4	70.3	62.9	51.9	36.0	24.2	48.9
-Monthly	30.6	37.2	53.0	62.7	70.0	76.0	80.9	81.3	73.1	61.2	46.1	35.1	58.9
-Monthly Dewpt.	20.2	28.3	41.9	49.7	55.8	60.7	64.5	65.0	57.5	51.3	37.9	23.4	46.4
Extremes													
-Highest	56	65	81	83	104	96	98	101	102	82	75	65	104
-Date	29	22	28	18	30	8	31	4	1	12	9	9	MAY 30
-Lowest	1	4	29	38	49	53	63	61	37	39	18	8	1
-Date	20	1	5	1	14	13	6	25	30	31	30	13	JAN 20
DEGREE DAYS BASE 65 °F:													
Heating	1059	773	377	108	10	0	0	0	63	146	562	921	4019
Cooling	0	0	12	43	172	336	501	512	313	38	0	0	1927
% OF POSSIBLE SUNSHINE													
	57	47	56	81	75	72	90	89	86	74	46	74	72
AVG. SKY COVER (tenths)													
Sunrise - Sunset	6.1	6.3	6.8	4.9	5.4	5.6	4.0	4.0	4.8	6.3	7.6	4.0	5.5
Midnight - Midnight	5.4	5.6	6.5	4.8	5.1	5.6	3.6	3.8	4.5	5.7	7.4	4.1	5.2
NUMBER OF DAYS:													
Sunrise to Sunset													
-Clear	9	7	9	11	12	8	18	17	12	8	5	16	132
-Partly Cloudy	6	7	4	10	9	10	5	8	9	6	6	7	87
-Cloudy	16	14	18	9	10	12	8	6	9	17	19	8	146
Precipitation													
.01 inches or more	4	6	12	5	3	8	5	7	7	11	8	4	80
Snow, Ice pellets													
1.0 inches or more	0	1	0	0	0	0	0	0	0	0	0	2	3
Thunderstorms	0	1	4	5	3	10	3	7	4	4	4	0	45
Heavy Fog, visibility													
1/4 mile or less	2	2	2	2	0	0	0	0	0	3	7	2	20
Temperature °F													
-Maximum													
90° and above	0	0	0	0	4	10	21	22	10	0	0	0	67
32° and below	6	6	0	0	0	0	0	0	0	0	1	7	20
-Minimum													
32° and below	28	16	4	0	0	0	0	0	0	0	9	25	82
0° and below	0	0	0	0	0	0	0	0	0	0	0	0	0
AVG. STATION PRESS. (mb)													
	975.6	973.6	969.2	967.8	966.1	968.2	969.2	969.2	969.9	970.5	969.9	975.8	970.5
RELATIVE HUMIDITY (%)													
Hour 00	72	77	73	73	72	69	68	69	70	81	79	72	73
Hour 06 (Local Time)	75	80	81	79	80	75	75	78	80	85	83	77	79
Hour 12	63	70	61	58	52	51	49	52	53	64	69	56	58
Hour 18	59	64	58	53	52	50	48	48	49	65	71	58	56
PRECIPITATION (inches):													
Water Equivalent													
-Total	0.92	3.71	6.60	5.35	1.49	8.34	1.33	2.63	4.59	5.23	3.73	0.26	44.18
-Greatest (24 hrs)	0.67	1.90	2.88	2.20	1.01	4.55	0.60	1.39	1.88	2.57	1.59	0.18	4.55
-Date	26-27	22-23	19-20	29	12-13	4-5	15-16	4-5	13	17-18	13-14	12-13	JUN 4-5
Snow, Ice pellets													
-Total	1.5	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	T	2.9	6.7
-Greatest (24 hrs)	0.8	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	T	2.8	2.8
-Date	27	4									30	12-13	DEC 12-13
WIND:													
Resultant													
-Direction (!!!)	340	074	135	154	155	168	152	141	153	120	001	349	141
-Speed (mph)	5.4	1.8	2.0	4.5	3.6	5.4	5.4	5.3	5.4	1.0	1.9	3.3	1.9
Average Speed (mph)	12.4	12.5	12.5	11.5	11.0	11.8	9.6	11.0	12.5	11.4	11.4	10.9	11.5
Fastest Obs. 1 Min.													
-Direction (!!!)	36	34	20	31	21	35	19	18	01	36	32	36	31
-Speed (mph)	43	31	30	46	29	33	31	28	32	37	30	31	46
-Date	19	26	3	4	13	10	24	4	13	4	19	11	APR 4
Peak Gust													
-Direction (!!!)	N	N	SW	NW	SE	N	S	W	N	N	S	N	NW
-Speed (mph)	51	40	62	66	51	53	46	51	44	46	45	38	66
-Date	19	26	3	4	12	10	24	7	22	4	8	23	APR 4

See Reference Notes on Page 68
Page 2

NORMALS, MEANS, AND EXTREMES

OKLAHOMA CITY, OKLAHOMA

LATITUDE: 35°24'N LONGITUDE: 97°36'W ELEVATION: FT. GRND 1285 BARO 01283 TIME ZONE: CENTRAL WBAN: 136

	1a)	JAN	FEB	MAR	APR	MAY	JUNE	JULY	AUG	SEP	OCT	NOV	DEC	YEAR
TEMPERATURE °F:														
Normals														
-Daily Maximum		46.6	52.2	61.0	71.7	79.0	87.6	93.5	92.8	84.7	74.3	59.9	50.7	71
-Daily Minimum		25.2	29.4	37.1	48.6	57.7	66.3	70.6	69.4	61.9	50.2	37.6	29.1	46
-Monthly		35.9	40.8	49.0	60.2	68.3	77.0	82.1	81.1	73.3	62.3	48.8	39.9	59
Extremes														
-Record Highest	32	79	84	93	100	104	105	108	110	102	96	87	86	1
-Year		1967	1981	1967	1972	1985	1980	1980	1980	1985	1972	1980	1955	AUG 19
-Record Lowest	32	-4	-3	3	20	37	47	53	51	37	22	11	-3	
-Year		1959	1979	1960	1957	1981	1954	1971	1956	1985	1957	1959	1983	JAN 19
NORMAL DEGREE DAYS:														
Heating (base 65°F)		902	678	506	184	41	0	0	0	15	145	486	778	37
Cooling (base 65°F)		0	0	13	40	147	360	530	499	264	61	0	0	19
% OF POSSIBLE SUNSHINE														
	31	59	60	63	66	67	74	79	79	72	69	60	59	1
MEAN SKY COVER (tenths)														
Sunrise - Sunset	37	5.9	5.8	5.9	5.7	5.8	5.0	4.4	4.3	4.6	4.6	5.1	5.5	5
MEAN NUMBER OF DAYS:														
Sunrise to Sunset														
-Clear	37	10.3	9.1	9.6	9.5	8.9	10.8	14.5	14.9	13.6	14.3	11.9	11.7	139
-Partly Cloudy	37	6.1	6.9	8.0	7.7	9.8	10.5	9.4	9.6	8.0	6.9	6.7	6.1	95
-Cloudy	37	14.6	12.2	13.4	12.7	12.2	8.6	7.1	6.5	8.4	9.8	11.3	13.2	130
Precipitation														
.01 inches or more	46	5.5	6.3	7.2	7.8	10.0	8.5	6.4	6.4	6.7	6.4	5.2	5.3	81
Snow, Ice pellets														
1.0 inches or more	46	0.9	1.0	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.6	3
Thunderstorms	46	0.5	1.4	3.0	5.5	9.1	8.7	6.2	6.3	4.5	3.1	1.3	0.6	50
Heavy Fog Visibility														
1/4 mile or less	37	4.0	3.3	1.9	1.0	0.6	0.4	0.3	0.3	0.8	1.6	2.1	3.3	19
Temperature of														
-Maximum														
90° and above	20	0.0	0.0	0.2	0.3	1.7	11.4	22.9	22.8	9.4	0.9	0.0	0.0	69
32° and below	20	6.2	2.6	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	2.5	11
-Minimum														
32° and below	20	23.8	17.6	8.0	1.1	0.0	0.0	0.0	0.0	0.0	0.6	8.6	20.8	80
0° and below	20	0.4	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0
AVG. STATION PRESS. (mb)														
	13	973.6	971.8	967.4	967.6	966.5	967.7	969.4	969.6	970.3	971.1	971.3	972.4	969
RELATIVE HUMIDITY (%)														
Hour 00	20	73	72	68	69	76	76	70	70	74	72	74	72	7
Hour 06	20	78	78	76	77	82	83	79	79	82	79	79	77	7
Hour 12 (Local Time)	20	60	57	53	52	57	56	49	49	54	52	56	58	5
Hour 18	20	60	54	49	49	54	54	45	45	52	54	60	61	5
PRECIPITATION (inches):														
Water Equivalent														
-Normal		0.96	1.29	2.07	2.91	5.50	3.87	3.04	2.40	3.41	2.71	1.53	1.20	30.86
-Maximum Monthly	46	5.68	3.71	7.16	10.78	12.07	9.94	8.44	6.77	9.64	13.18	5.46	8.14	13.18
-Year		1949	1985	1948	1947	1982	1979	1959	1966	1970	1983	1964	1984	OCT 1981
-Minimum Monthly	46	T	T	T	0.62	0.33	0.63	T	0.25	T	T	T	0.03	T
-Year		1976	1947	1940	1971	1942	1952	1983	1978	1948	1958	1949	1955	JUL 1981
-Maximum in 24 hrs	46	3.10	2.21	3.44	3.80	5.63	4.55	5.75	3.34	7.68	8.95	2.01	2.55	8.95
-Year		1982	1978	1944	1970	1970	1985	1981	1966	1970	1963	1961	1984	OCT 1981
Snow, Ice pellets														
-Maximum Monthly	46	17.3	12.0	13.9	0.7						T	7.5	8.2	17.3
-Year		1949	1978	1968	1957						1967	1972	1960	JAN 1949
-Maximum in 24 hrs	46	8.0	6.1	6.4	0.7						T	5.5	5.7	8.4
-Year		1944	1979	1948	1957						1967	1972	1984	MAR 1948
WIND:														
Mean Speed (mph)	37	12.9	13.3	14.6	14.5	12.7	12.3	10.9	10.6	11.2	12.0	12.4	12.6	12.9
Prevailing Direction through 1963		N	N	SSE	SSE	SSE	SSE	SSE	SSE	SSE	SSE	S	S	SSE
Fastest Mile														
-Direction (!!!)	30	W	W	S	NW	SE	NNW	NW	S	NW	S	S	NW	NW
-Speed (MPH)	30	63	61	61	75	72	87	73	56	54	65	66	56	87
-Year		1960	1960	1954	1960	1960	1951	1960	1952	1961	1951	1958	1961	JUN 1951
Peak Gust														
-Direction (!!!)	2	N	N	SW	NW	SE	N	S	W	N	W	NW	N	NW
-Speed (mph)	2	51	46	62	66	51	53	46	51	44	50	47	40	66
-Date		1985	1984	1985	1985	1985	1985	1985	1985	1985	1984	1984	1984	APR 1985

!!! See Reference Notes on Page 68.
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PRECIPITATION (inches)

OKLAHOMA CITY, OKLAHOMA

YEAR	JAN	FEB	MAR	APR	MAY	JUNE	JULY	AUG	SEP	OCT	NOV	DEC	ANN
1956	0.26	1.31	0.56	2.43	5.07	2.46	2.19	0.86	0.20	3.76	1.49	1.97	22.1
1957	1.49	0.86	2.19	7.32	8.67	8.60	0.63	1.08	5.58	2.65	2.25	0.72	42.1
1958	1.29	0.83	3.39	2.22	3.10	7.17	2.02	3.56	3.87	T	0.60	1.11	29.1
1959	0.31	0.69	1.40	2.81	7.60	2.93	8.44	4.10	7.90	6.08	0.94	3.26	46.1
1960	1.47	2.12	0.85	3.01	6.79	5.68	7.69	2.89	1.62	4.66	0.24	2.48	39.1
1961	0.15	1.98	3.35	0.73	1.92	3.86	4.82	2.91	7.37	2.86	3.81	1.04	34.1
1962	1.45	1.02	0.80	2.16	2.64	7.84	1.71	2.26	3.08	2.43	1.34	0.76	27.1
1963	0.21	0.22	3.21	2.77	1.91	2.35	6.19	1.61	1.91	2.05	2.45	0.89	25.1
1964	0.83	2.17	1.30	2.06	5.21	0.77	2.01	4.91	2.96	0.84	5.46	0.62	29.1
1965	0.98	0.85	0.86	3.24	2.14	3.65	1.57	3.37	3.94	1.00	0.06	2.51	24.1
1966	1.05	2.39	1.30	3.68	0.88	2.63	2.38	6.77	2.82	0.37	0.84	0.45	25.1
1967	0.77	0.20	2.49	5.71	4.25	2.27	1.21	1.40	3.15	2.92	0.40	1.04	25.1
1968	2.19	1.02	2.84	3.03	8.40	2.39	1.41	3.75	2.64	2.40	4.11	1.33	35.1
1969	0.20	1.93	3.01	1.66	3.99	4.92	1.42	2.38	6.51	1.58	0.06	1.44	29.1
1970	0.32	0.29	2.09	5.33	6.53	2.45	1.30	0.80	9.64	3.29	1.03	0.26	33.1
1971	0.75	1.95	0.07	0.62	2.68	5.15	4.13	2.13	4.25	2.62	0.29	2.79	27.1
1972	0.21	0.43	1.13	3.10	4.03	1.36	3.22	1.82	2.04	7.17	2.28	0.84	27.1
1973	3.39	0.31	6.76	2.32	3.61	6.31	3.38	1.36	8.00	3.05	2.81	0.47	41.1
1974	0.10	2.68	3.12	4.66	5.01	3.36	0.48	4.42	6.24	5.57	2.34	1.47	39.1
1975	1.99	1.90	1.72	1.92	8.76	4.82	7.71	0.60	1.92	0.84	1.77	1.30	35.1
1976	T	0.33	3.09	2.94	4.36	0.88	1.38	1.46	1.53	1.78	0.12	0.19	18.0
1977	0.32	1.40	1.30	2.88	7.97	2.00	4.10	3.08	1.20	2.41	1.59	0.34	28.5
1978	1.26	3.23	1.32	1.65	10.12	4.04	3.75	0.25	0.96	1.02	2.88	0.70	31.1
1979	1.55	0.63	2.73	2.78	7.29	9.94	5.62	3.78	0.72	1.58	1.93	2.57	41.1
1980	1.69	1.29	1.38	2.16	9.00	2.52	0.42	0.60	2.21	0.99	0.51	1.58	24.3
1981	0.19	1.15	2.87	2.97	2.73	7.49	6.45	3.61	1.48	7.70	2.11	0.20	38.9
1982	3.68	0.98	1.63	1.92	12.07	4.06	2.11	1.13	2.86	1.03	2.78	1.94	36.1
1983	2.62	1.71	2.51	2.34	6.88	3.18	T	3.18	0.90	13.18	1.90	0.70	39.1
1984	0.35	1.16	4.70	1.79	1.62	3.48	0.30	2.35	1.01	6.64	2.05	8.14	33.5
1985	0.92	3.71	6.60	5.35	1.49	8.34	1.33	2.63	4.59	5.23	3.73	0.26	44.1
Record Mean	1.26	1.26	2.20	3.20	5.15	3.99	2.74	2.66	3.13	2.95	1.90	1.47	31.8

See Reference Notes on Page 68.
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AVERAGE TEMPERATURE (deg. F)

OKLAHOMA CITY, OKLAHOMA

YEAR	JAN	FEB	MAR	APR	MAY	JUNE	JULY	AUG	SEP	OCT	NOV	DEC	ANNUAL
1956	36.1	40.6	50.5	57.9	72.6	78.1	83.0	84.4	76.6	66.3	47.1	41.7	61.2
1957	33.2	45.4	46.7	55.9	65.7	74.2	83.5	80.7	69.3	57.3	45.6	44.3	58.9
1958	39.4	35.5	41.0	56.1	69.4	76.8	80.1	79.9	73.2	61.9	51.4	36.5	58.4
1959	33.0	40.0	49.6	58.5	70.4	75.7	77.4	80.1	72.6	58.3	43.5	44.2	58.6
1960	36.3	34.5	39.7	61.7	66.5	77.3	78.2	79.3	74.5	64.4	51.4	37.4	58.4
1961	36.3	42.7	52.2	59.2	67.7	74.1	79.1	78.4	70.6	63.0	46.5	36.7	58.4
1962	32.3	45.1	48.1	58.3	74.3	75.3	82.2	82.5	71.4	64.7	49.4	40.6	60.4
1963	28.3	40.0	53.8	64.8	70.3	78.7	83.1	82.0	75.1	71.1	52.3	33.0	61.1
1964	40.1	38.5	47.1	64.1	70.0	77.4	85.3	80.8	72.3	60.3	49.9	37.9	60.3
#1965	38.8	39.5	40.7	65.5	71.3	78.0	84.6	80.6	73.9	62.8	56.3	48.8	61.8
1966	33.8	38.6	52.8	57.8	68.7	77.9	86.4	78.8	70.5	60.9	54.3	37.8	59.9
1967	41.8	41.7	56.4	65.4	66.9	77.4	79.7	79.3	70.9	62.6	49.4	39.8	60.9
1968	36.6	36.4	50.7	58.1	64.7	74.8	79.8	80.0	70.5	62.1	46.2	38.1	58.1
1969	38.8	42.3	42.0	60.4	67.7	75.0	83.9	80.2	73.3	57.6	48.7	40.4	59.2
1970	31.8	42.7	45.0	60.2	69.2	76.3	82.1	83.6	75.1	57.9	46.0	43.9	59.5
1971	36.9	39.1	49.1	60.4	67.3	78.6	80.7	77.2	73.3	63.4	49.0	42.2	59.8
1972	34.9	42.1	53.4	63.2	67.6	79.0	79.8	80.4	75.8	61.1	43.4	34.4	59.6
1973	33.3	39.8	52.5	56.0	66.8	75.2	79.8	79.7	70.6	64.3	53.1	39.3	59.2
1974	35.0	44.4	54.8	60.0	71.5	74.1	82.7	78.5	65.5	63.5	49.3	39.6	59.9
1975	40.3	36.5	46.1	58.7	67.4	75.1	78.0	80.1	68.3	63.4	50.7	41.8	58.8
1976	39.0	52.2	52.4	61.6	63.6	74.8	79.8	81.3	72.6	56.5	43.9	38.8	59.7
1977	29.2	45.9	54.1	62.5	70.0	79.6	83.0	80.7	78.0	62.7	50.9	40.0	61.4
1978	26.3	29.4	49.1	64.5	68.1	77.3	87.0	82.6	79.7	64.7	50.4	36.9	59.7
1979	25.4	31.5	51.2	58.1	65.8	75.2	81.0	80.0	73.1	65.7	46.5	43.3	58.1
1980	38.2	38.2	46.3	56.7	69.0	81.4	88.3	88.0	76.3	61.1	50.3	41.9	61.3
1981	37.7	43.9	51.9	65.6	65.7	78.4	84.2	78.8	74.1	60.1	50.3	39.1	60.8
1982	35.3	37.7	52.7	57.5	68.2	72.2	81.0	84.1	74.5	62.7	48.6	43.2	59.8
1983	38.6	42.6	48.8	54.0	64.6	73.4	81.6	84.0	74.9	62.7	50.4	25.8	58.5
1984	34.0	45.4	46.4	56.5	68.4	78.6	81.6	82.6	71.5	61.6	49.7	43.0	59.9
1985	30.6	37.2	53.0	62.7	70.0	76.0	80.9	81.3	73.1	61.2	46.1	35.1	58.9
Record Mean	36.8	40.7	49.9	60.1	68.1	76.9	81.6	81.3	73.8	62.5	49.4	39.8	60.1
Max	46.8	51.5	61.3	71.2	78.2	87.2	92.4	92.4	84.8	73.7	60.0	49.6	70.8
Min	26.8	29.9	38.4	49.1	57.9	66.7	70.8	70.1	62.8	51.3	38.7	29.9	49.4

See Reference Notes on Page 68.
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HEATING DEGREE DAYS Base 65 deg. F

OKLAHOMA CITY, OKLAHOMA

SEASON	JULY	AUG	SEP	OCT	NOV	DEC	JAN	FEB	MAR	APR	MAY	JUNE	TOTAL
1956-57	0	0	0	72	531	713	978	542	560	264	72	9	3761
1957-58	0	0	17	263	572	633	789	818	737	268	24	3	4124
1958-59	0	0	17	158	410	876	983	692	481	248	27	0	3899
1959-60	0	0	24	206	640	644	880	877	777	134	73	0	4255
1960-61	0	0	2	96	407	852	885	618	395	247	49	0	3551
1961-62	0	0	40	106	547	868	1006	560	524	230	5	0	3886
1962-63	0	0	19	129	458	750	1131	692	372	101	50	0	3702
1963-64	0	0	7	23	381	988	761	762	547	116	27	6	3618
1964-65	0	0	24	150	454	831	808	706	745	71	2	0	3791
1965-66	0	0	28	129	262	496	961	734	388	223	62	0	3283
1966-67	0	0	6	166	338	837	713	647	307	71	77	0	3162
1967-68	0	0	27	155	464	773	872	826	444	215	71	0	3847
1968-69	0	0	0	152	561	829	808	629	708	158	38	2	3885
1969-70	0	0	0	274	481	752	1022	620	615	187	31	12	3994
1970-71	0	0	18	254	559	651	866	718	492	163	36	0	3757
1971-72	0	0	59	88	475	702	923	660	365	144	46	0	3462
1972-73	0	0	23	225	640	940	975	701	380	283	55	0	4222
1973-74	0	0	37	99	362	787	922	573	330	168	8	0	3286
1974-75	0	0	56	88	463	784	763	792	583	235	29	0	3793
1975-76	0	0	64	126	430	713	801	367	406	128	100	0	3135
1976-77	0	0	19	306	629	805	1103	529	338	107	7	0	3843
1977-78	0	0	0	115	420	766	1192	990	493	90	64	0	4130
1978-79	0	0	2	89	437	866	1221	932	434	217	81	0	4279
1979-80	0	0	2	92	551	669	823	771	572	249	24	0	3753
1980-81	0	0	23	180	444	710	839	587	400	69	69	0	3321
1981-82	0	0	22	189	434	797	913	759	382	248	25	13	3782
1982-83	0	0	14	156	490	671	809	622	496	345	96	9	3708
1983-84	0	0	25	117	439	1207	955	561	572	263	45	0	4184
1984-85	0	0	75	162	462	676	1059	773	377	108	10	0	3702
1985-86	0	0	63	146	562	921							

See Reference Notes on Page 6B.
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COOLING DEGREE DAYS Base 65 deg. F

OKLAHOMA CITY, OKLAHOMA

YEAR	JAN	FEB	MAR	APR	MAY	JUNE	JULY	AUG	SEP	OCT	NOV	DEC	TOTAL
1969	0	0	0	29	128	310	593	477	255	52	0	0	1844
1970	0	0	1	47	169	357	536	582	328	38	0	0	2058
1971	0	0	4	31	117	416	493	388	313	45	3	0	1810
1972	0	0	11	97	133	429	470	483	351	109	0	0	2085
1973	0	0	0	19	119	312	465	462	216	83	1	0	1681
1974	0	0	22	26	217	280	553	426	80	47	0	0	1681
1975	0	0	1	53	108	310	410	476	170	83	4	0	1681
1976	0	1	23	33	62	300	468	512	253	50	0	0	1681
1977	0	1	8	37	170	445	565	491	395	49	0	0	1681
1978	0	0	8	80	165	378	690	553	450	87	0	0	1681
1979	0	0	10	18	112	314	505	471	252	121	0	0	1681
1980	0	0	0	7	155	498	729	721	366	65	0	0	1681
1981	0	4	0	94	98	409	603	435	304	41	0	0	1681
1982	0	0	9	28	130	234	503	508	305	90	0	0	1681
1983	0	0	0	20	91	266	523	599	329	54	0	0	1681
1984	0	0	10	16	159	414	521	551	279	64	0	0	1681
1985	0	0	12	43	172	336	501	512	313	38	0	0	1681

See Reference Notes on Page 6B.
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SNOWFALL (inches)

OKLAHOMA CITY, OKLAHOMA

SEASON	JULY	AUG	SEP	OCT	NOV	DEC	JAN	FEB	MAR	APR	MAY	JUNE	TOTAL
1956-57	0.0	0.0	0.0	0.0	0.0	1.6	T	T	0.1	0.7	0.0	0.0	2.4
1957-58	0.0	0.0	0.0	T	T	0.0	1.4	0.5	6.8	0.0	0.0	0.0	8.7
1958-59	0.0	0.0	0.0	0.0	2.6	7.0	3.3	0.5	1.2	0.0	0.0	0.0	14.6
1959-60	0.0	0.0	0.0	0.0	0.3	T	8.2	5.4	3.2	0.0	0.0	0.0	17.1
1960-61	0.0	0.0	0.0	0.0	0.0	8.2	1.9	7.2	0.7	T	0.0	0.0	18.0
1961-62	0.0	0.0	0.0	0.0	0.0	0.8	8.5	0.8	0.0	0.0	0.0	0.0	10.1
1962-63	0.0	0.0	0.0	0.0	T	1.2	0.5	1.6	0.0	0.0	0.0	0.0	3.3
1963-64	0.0	0.0	0.0	0.0	0.0	2.5	T	0.6	1.5	0.0	0.0	0.0	4.6
1964-65	0.0	0.0	0.0	0.0	0.0	T	2.0	0.9	0.1	0.0	0.0	0.0	3.0
1965-66	0.0	0.0	0.0	0.0	0.0	T	5.8	2.3	0.0	0.0	0.0	0.0	8.1
1966-67	0.0	0.0	0.0	0.0	T	0.6	T	0.1	0.4	0.0	0.0	0.0	1.1
1967-68	0.0	0.0	0.0	T	1.2	1.2	0.4	7.7	13.9	0.0	0.0	0.0	24.4
1968-69	0.0	0.0	0.0	0.0	1.4	1.3	T	1.3	8.2	0.0	0.0	0.0	12.2
1969-70	0.0	0.0	0.0	0.0	0.0	4.2	0.4	T	2.3	T	0.0	0.0	6.9
1970-71	0.0	0.0	0.0	0.0	T	T	T	5.1	0.7	0.0	0.0	0.0	5.8
1971-72	0.0	0.0	0.0	0.0	0.5	5.2	0.8	4.9	0.0	0.0	0.0	0.0	11.4
1972-73	0.0	0.0	0.0	0.0	7.5	1.4	8.3	0.2	T	T	0.0	0.0	17.4
1973-74	0.0	0.0	0.0	0.0	0.0	0.6	0.7	1.0	0.5	0.0	0.0	0.0	2.8
1974-75	0.0	0.0	0.0	0.0	1.0	2.0	0.6	0.9	0.1	0.0	0.0	0.0	4.6
1975-76	0.0	0.0	0.0	0.0	0.7	3.9	T	0.3	0.0	0.0	0.0	0.0	4.9
1976-77	0.0	0.0	0.0	0.0	0.3	T	2.8	0.4	0.0	0.0	0.0	0.0	3.5
1977-78	0.0	0.0	0.0	0.0	T	0.0	8.4	12.0	T	0.0	0.0	0.0	20.4
1978-79	0.0	0.0	0.0	0.0	0.0	3.3	4.0	6.1	0.0	0.0	0.0	0.0	13.4
1979-80	0.0	0.0	0.0	0.0	T	T	T	1.8	T	0.0	0.0	0.0	1.8
1980-81	0.0	0.0	0.0	0.0	4.0	0.0	T	T	0.0	0.0	0.0	0.0	4.0
1981-82	0.0	0.0	0.0	0.0	0.0	T	1.0	3.9	2.5	0.0	0.0	0.0	7.4
1982-83	0.0	0.0	0.0	0.0	T	T	5.1	4.3	T	0.0	0.0	0.0	9.4
1983-84	0.0	0.0	0.0	0.0	T	1.9	5.6	2.0	T	0.0	0.0	0.0	9.5
1984-85	0.0	0.0	0.0	0.0	T	6.1	1.5	2.3	0.0	0.0	0.0	0.0	9.9
1985-86	0.0	0.0	0.0	0.0	T	2.9							
Record													
Mean	0.0	0.0	0.0	T	0.5	1.6	2.9	2.5	1.5	T	0.0	0.0	9.0

See Reference Notes on Page 6B.
Page 6A

REFERENCE NOTES

OKLAHOMA CITY, OKLAHOMA

GENERAL

T - TRACE AMOUNT.
BLANK ENTRIES DENOTE MISSING/UNREPORTED DATA.
INDICATES A STATION OR INSTRUMENT RELOCATION.
SEE STATION LOCATION TABLE ON PAGE 8.

SPECIFIC

PAGE 2
PM - INCLUDES LAST DAY OF PREVIOUS MONTH

PAGE 3

1st - LENGTH OF RECORD IN YEARS, ALTHOUGH
INDIVIDUAL MONTHS MAY BE MISSING.
* LESS THAN .05
NORMALS - BASED ON THE 1951-1980 RECORD PERIOD.
EXTREMES - DATES ARE THE MOST RECENT OCCURRENCE
WIND DIR. - NUMERALS SHOW TENS OF DEGREES CLOCKWISE
FROM TRUE NORTH. "00" INDICATES CALM.
RESULTANT DIRECTIONS ARE GIVEN TO WHOLE DEGREES.

EXCEPTIONS

PAGE 3

1. FASTEST MILE WIND IS THROUGH OCTOBER 1981.
PAGES 4A, 4B, 6A

RECORD MEANS ARE THROUGH THE CURRENT YEAR,
BEGINNING IN 1891 FOR TEMPERATURE
1891 FOR PRECIPITATION
1940 FOR SNOWFALL

OKLAHOMA CITY, OKLAHOMA

Oklahoma City is located along the North Canadian River, a frequently nearly-dry stream, at the geographic center of the state. It is not quite 1,000 miles south of the Canadian Border and a little less than 500 miles north of the Gulf of Mexico. The surrounding country is gently rolling with the nearest hills or low mountains, the Arbuckles, 80 miles south. The elevation ranges around 1,250 feet above sea level.

Although some influence is exerted at times by warm, moist air currents from the Gulf of Mexico, the climate of Oklahoma City falls mainly under continental controls characteristic of the Great Plains Region. The continental effect produces pronounced daily and seasonal temperature changes and considerable variation in seasonal and annual precipitation. Summers are long and usually hot. Winters are comparatively mild and short.

During the year, temperatures of 100 degrees or more occur on an average of 10 days, but have occurred on as many as 50 days or more. While summers are usually hot, the discomforting effect of extreme heat is considerably mitigated by low humidity and the prevalence of a moderate southerly breeze. Approximately one winter in three has temperatures of zero or lower.

The length of the growing season varies from 180 to 251 days. Average date of last freeze is early April and average date of first freeze is early November. Freezes have occurred in early October.

During an average year, skies are clear approximately 40 percent of the time, partly cloudy 25 percent, and cloudy 35 percent of the time. The city is almost smoke-free as a result of favorable atmospheric conditions and the almost exclusive use of natural gas for heating. Flying conditions are generally very good with flight by visual flight rules possible about 96 percent of the time.

Summer rainfall comes mainly from showers and thunderstorms. Winter precipitation is generally associated with frontal passages. Measurable precipitation has occurred on as many as 122 days and as few as 55 days during the year. The seasonal distribution of precipitation is normally 12 percent in winter, 34 percent in spring, 30 percent in summer, and 24 percent in fall. The period with the least number of days with precipitation is November through January, and the month with the most rainy days is May. Thunderstorms occur most often in late spring and early summer. Large hail and/or destructive wind on occasion accompany these thunderstorms.

Snowfall averages less than 10 inches per year and seldom remains on the ground very long. Occasional brief periods of freezing rain and sleet storms occur.

Heavy fogs are infrequent. Prevailing winds are southerly except in January and February when northerly breezes predominate.

STATION LOCATION

OKLAHOMA CITY, OKLAHOMA

Location	Occupied from	Occupied to	Airline distance and direction from previous location	Latitude North	Longitude West	Elevation above								Remarks
						Sea level	Ground							
							Ground at tem-perature site	Wind instruments	Extreme thermometers	Psychrometer	Sunshine Switch	Tipping bucket rain gage	Weighting rain gage	
Automatic Observing Equipment *														
* Type M = AMOS T = AUTOM														
<u>CITY</u>														
Overholser Opera House SE Corner, Robinson and Grand	11/01/90	6/30/02		35° 28'	97° 30'	1196	53	54	54		43		43	
Culbertson Building SE Corner, Broadway and Grand	7/01/02	1/16/06	400 feet E	35° 28'	97° 30'	1196	86	79	79		71		71	
WB Observatory 1923 Classen Boulevard	1/17/06	3/01/55	1.9 mile NW	35° 29'	97° 32'	1254		10				5		
Offices consolidated at Airport 4/1/54.														
<u>AIRPORT</u>														
Administration Building Will Rogers Field 7 miles SW of PO	4/02/32	5/20/54		35° 24'	97° 36'	1280	70	33	31		4	4	4	
Page Hanger #3 Will Rogers Field	5/21/54	10/21/65	3/16 mile N	35° 24'	97° 36'	1280	55	7	6		4	5	4	a5
Weather Bureau Building Will Rogers World AP	10/21/65	Present	200 feet NE	35° 24'	97° 36'	1285	20	7 c5	6 c5	122	4	5	4	b5 NA
a = Telepsychrometer (6') 5/21/54-12/31/60. Hygro. comm. 12/31/60. b = 1 mile south of previous site. c = Effective December 1975. 1 = Commissioned 10/20/65.														

Offices consolidated at Airport 4/1/54.

a - Telepsychrometer(6'')5/21/54-12/31/60, Hygro. comm. 12/31/60.
b - 1 mile south of previous site.
c - Effective December 1975.
2 - Commissioned 10/20/65.

SUBSCRIPTION:
Price and ordering information available through: National Climatic Data Center, Federal Building, Asheville, North Carolina 28801

USCOM-NOAA-ASHEVILLE, N.C. - 620

U. S. DEPARTMENT OF COMMERCE
NATIONAL CLIMATIC DATA CENTER
FEDERAL BUILDING
ASHEVILLE, N.C. 28801

AN EQUAL OPPORTUNITY EMPLOYER

POSTAGE AND FEES PAID
U.S. DEPARTMENT OF COMMERCE
COM 210



FIRST CLASS



The
University of Oklahoma

DEPARTMENT OF PHYSICAL PLANT
Office of the Director
160 Feiga Street, Room 202
Norman, Oklahoma 73019

August 22, 1984

Mr. Richard A. Moore
Project Manager
Metcalf & Eddy, Inc.
50 Staniford Street
Boston, Massachusetts 02114

NEXT GENERATION WEATHER RADAR SYSTEM, OKLAHOMA UNIVERSITY NORTH CAMPUS, MAX
WESTHEIMER FIELD, CLEVELAND COUNTY, OKLAHOMA

Please refer to your letter dated June 18, 1984.

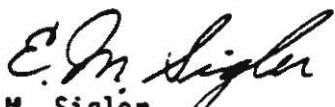
The location referred to in your letter is located on our North Campus. Electric power is purchased from Oklahoma Gas and Electric Company at 7200/12470Y, three-phase system. At this time, we are building a new 2500 KVA, three-phase substation to replace our old one. Oklahoma University distributes power at 2400/4160Y volts, three-phase.

We propose to install a "riser" on existing pole and install an underground cable to a three-phase, 225 KVA padmount transformer located adjacent to your office. Secondary of this transformer can be 120/208Y volts or 277/480Y volts to suit your requirements. We prefer L.V. metering.

Our present method of billing for electric power is based on what we pay OG&E plus a small adder of 7%. We buy in bulk rate which presently is 0.046 cents per KWH. Naturally by 1988, OG&E charges will be different.

Southwestern Bell Telephone System serves the area with telephones.

If additional information is desired, please feel free to contact me.


E. M. Sigler
O.U. Physical Plant Engineering

NEXRAD

Electric Utility Questionnaire

SITE NAME: Oklahoma City, Oklahoma

LOCATION: Max Westheimer Field, Norman, Oklahoma

A new service of 400 amperes at 208Y/120 volts, 3 phase, 60 hertz, 4 wire is planned. It is currently expected that the increased service will be required by January 1988.

1. What is the capacity of the existing transformers? KVA Will install to yours
Sec. Volt Requirement stated
Phases above, 150 KVA or 225 KVA
2. What is primary voltage? Volts 4.16 or 12.5 KV
Phases 3
3. What is service type? Aboveground From OG&E
Underground Proposed OU's system
4. Can required service be obtained from existing system? Yes X
No
5. Can power be purchased at primary voltage? Yes
No X
6. What approximate costs may be incurred by the customer? (Identify major items.) Primary Metered \$
Secondary Metered \$ None
7. What are applicable rates? (Provide copies.) Primary Metered \$
Secondary Metered \$.0494887
per KWH as of 11-15-85
8. What is the quality of service?

	1980	1981	1982	1983	1984
a. Number of outages or breaker operations	2	3	1	2	1
b. Duration (in days) of longest outage	5 hours, lost substation transformer				
c. Incidence of excessive voltage	None				
d. Incidence of low voltage	None				
9. Provide any additional pertinent general information which may be a factor in increasing this service. See letter



Southwestern Bell
Telephone

121 Dear A McGee Avenue
Room 275
Oklahoma City, Oklahoma 73102-6

November 6, 1985

Metcalf & Eddy Inc.
Attn: Mark Johnson
10 Harvard Mill Square
Wakefield, MA 01888-4043

Dear Mark:

This is a reply to your questionnaire of October 25, 1985. So that you may follow the continuity of your questionnaire, I will provide answers to your questions as you have asked them.

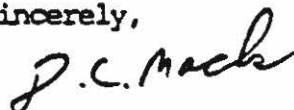
1. The location of our central office is 101 S. Webster, Norman, Oklahoma. It is approximately 1.75 miles to the South-Southeast of the NWS proposed facility.
2. The distance from the nearest terminal to the NWS facility is not available, as the facility has not been completed; therefore, engineering, installation, and other technical considerations, which would decide the terminals used, are unknown.
3. The number of cable pairs between the central office and the facility, as it would apply to service provided to the NWS site, is difficult to determine, given the variable of time, change in the number of customers in the same area over the next couple of years, etc. The exact number of pairs currently in place at present is classified and cannot be given.
4. We do provide 56 KBPS dataphone service in the Norman area at this time, and foresee this service continuing to be offered. The site at Max Westheimer Field may be provided with this service.
5. The number of data grade lines installed at the site currently is classified information, and further installation would be dependent upon customer demand.
6. 9600BPS service is within our technological ability to provide the NWS site with, according to our engineers. However, once again the number of lines, the costs involved to lay any needed additional lines are items which are classified, or depend upon variables impossible to forecast accurately.

8. I am sorry that much of this information is so general, but I'm sure you can appreciate the uncertainty of forecasting future events or requirements. I can assure you that we will do whatever is necessary to provide top-grade service for a facility which is so important to the lives of our customers, and Southwestern Bell employees.

Any further inquiries may be made to:

Southwestern Bell Telephone Company
Attn: Dana Mack, Position 328
121 Dean A. McGee, Room 275
Oklahoma City, OK 73102

Sincerely,

A handwritten signature in cursive script that reads "D.C. Mack".

Service Representative

Operational Support Facility
Norman, OK
Rev. 0

Appendix C

LOCAL POINTS OF CONTACT

Appendix C

LOCAL POINTS OF CONTACT

City of Norman

Gary Allen, Assistant Director of Code Enforcement/Building Official
(405) 321-1330, Ext. 25

City of Norman
Code Enforcement Department
P.O. Box 370
201 West Gray Street, Building C
Norman, OK 73070

University of Oklahoma

John E. Sexton, Director (405) 325-2138
Real Estate Development
University of Oklahoma
1000 Asp Ave., Room 210
Norman, OK 73019

W. E. Pack, Electrical Engineer (405) 325-6411
University of Oklahoma
Department of Physical Plant
160 Felgar Street, Room 202
Norman, OK 73019

Mike Moorman, Campus Planner (405) 325-6006
University of Oklahoma
Architectural and Engineering Services
1652 Cross Center Drive
Norman, OK 73019

National Weather Service

Kenneth C. Crawford, Meteorologist-In-Charge (405) 686-4155
National Oceanic & Atmospheric Administration
NWS Forecast Office
P.O. Box 59997
Oklahoma City, OK 73159

Cal Phipps (arranged telephone service) (817) 334-2655
National Weather Service--Southern Region
10A29 Federal Office Building
819 Taylor Street
Fort Worth, TX 76102

Telephone Service

Kathy Henderson, Communications Consultant (405) 278-5401
Southwestern Bell Telephone Company
111 Dean A. McGee, Room 1151
Oklahoma City, OK 73102

Environmental Information

Dr. Gary Snell, Director (405) 325-4034
Oklahoma Biological Survey
University of Oklahoma
Sutton Hall, Room 303
625 Elm Street
Norman, OK 73019

Architect

Dwaine Rhodes, Architect (405) 321-1020
Kaighn Associates Architects, Inc.
111 North Peters, Suite 500
Norman, OK 73069

Operational Support Facility
Norman, OK
Rev. 0

Appendix D

ECAC FREQUENCY ASSIGNMENT ANALYSES

**DEPARTMENT OF DEFENSE
Electromagnetic Compatibility Analysis Center
Annapolis, Maryland 21402**

Rec'd
JUN 12 1986

NEXRAD OSF SITE FREQUENCY ANALYSIS, NORMAN, OK

Prepared for

NEXRAD Joint System Program Office
National Weather Service/Code WX7
8060 13th Street
Silver Spring, MD 20910



MAY 1986

Prepared by
Larrey A. Snyder

**IIT Research Institute
Under Contract to
Department of Defense**

Distribution limited to US Government agencies only; Test and Evaluation; May 1986. Other requests for this document must be referred to NEXRAD JSPO.

NEXRAD OSF SITE FREQUENCY ANALYSIS

Date analysis completed: May 86

Using GMF data dated: Mar 86

I. BACKGROUND

The National Severe Storms Laboratory (NSSL) in Norman, OK has been chosen as the location of the NEXRAD Operational Support Facility (OSF). The OSF will act as a research and development facility to support test and evaluation of the NEXRAD system in its initial deployment. Testing of new weather algorithms and procedures will be performed at the facility. The OSF will also serve as a training facility for NEXRAD technical personnel.

A study performed by NTIA ¹, measurements made by SRI International², and a survey of the ECAC data base show that the 2700-2900 MHz band in the Norman, OK area is heavily congested. The problem of identifying candidate operating frequencies in this environment is compounded since a network NEXRAD will be deployed at nearby Twin Lakes.

During the In-depth Site Survey process, conducted by SRI International, measurements of received power levels throughout the 2700-3000 MHz band were made at both the OSF site and the Twin Lakes site. Based on results of the measurements and Preliminary Site Survey studies, SRI has recommended that the OSF at Norman be assigned a frequency in the 2900-3000 MHz band and the Twin Lakes NEXRAD be assigned a frequency in the range of 2885-2895 MHz².

Since the OSF will be a test and development facility for NEXRAD, the Joint System Program Office (JSPO) has requested that an assignment for the OSF be found in the 2700-2900 MHz band, if possible. In response to the JSPO request, a frequency assignment analysis was performed for each of the above sites.

II. PURPOSE

The purpose was to analyze the existing frequency assignments in the vicinity of the NEXRAD OSF site in Norman, OK and recommend a primary and secondary operational frequency for NEXRAD. If possible, the primary frequency recommended should be in the 2700-2900 MHz band.

III. APPROACH

A frequency assignment analysis was performed for the NEXRAD OSF at Norman, OK (latitude 35-14-09 N, longitude 97-27-47 W) to select candidate frequencies for NEXRAD use. The ECAC data base was used to identify environmental equipments and respective operating frequencies in the 2700-3000 MHz band. The select was performed for equipments within 150 miles of the proposed site, using the Government Master File (GMF) data as of Mar 1986. The total number of assignments identified was 80. Histograms of the number of frequency assignments in each 5 MHz of spectrum and recommendations from SRI International were used to determine candidate frequencies to be analyzed. The following candidate frequencies were selected:

2705 MHz

2745 MHz

2845 MHz

2890 MHz

2950 MHz

IV. ANALYSIS

For each candidate frequency, analyses were performed for NEXRAD transmitter vs. environmental receivers and environmental transmitters vs. NEXRAD receiver. Data for environmental equipment was obtained from the GMF and from technical manuals. A 9 dB interference-to-noise threshold $(I/N)_t$ criterion discussed in Section 2 of Reference 3 was used.

Predicted results for each potential NEXRAD frequency follows:

NEXRAD tuned to 2705 MHz - Two possible interactions were identified. Plots of I/N are included in Enclosure D1.

1. ASR-4 (Will Rogers) @ 2725 MHz vs. NEXRAD RX
2. WDR-73 (NSSL) @ 2850 MHz vs. NEXRAD RX

NEXRAD tuned to 2745 MHz - Four possible interactions were identified. Plots of I/N are included in Enclosures D2-D3.

1. NEXRAD TX vs. ASR-8 (Will Rogers) @ 2750 MHz.
2. ASR-8 TX (Will Rogers) @ 2750 MHz vs. NEXRAD RX
3. WSR-57 TX (NSSL) @ 2834 MHz vs. NEXRAD RX
4. WDR-73 TX (NSSL) @ 2850 MHz vs. NEXRAD RX

NEXRAD tuned to 2845 MHz - Five possible interactions were identified. Plots of I/N are included in Enclosures D4-D5.

1. NEXRAD TX vs. WSR-57 RX (NSSL) @ 2834 MHz
2. NEXRAD TX vs. WDR-73 RX (NSSL @ 2850 MHz
3. WSR-57 TX (NSSL) @ 2834 MHz vs. NEXRAD RX
4. WDR-73 TX (NSSL) @ 2850 MHz vs. NEXRAD RX
5. ASR-8 TX (Will Rogers) @ 2835 MHz vs. NEXRAD RX

NEXRAD tuned to 2890 MHz - This frequency was chosen assuming the WSR-57 at Will Rogers Airport will be decommissioned upon installation of the NEXRAD radar at Twin Lakes, OK. Two possible interactions were identified. Plots of I/N are included in Enclosures D6-D7.

1. WDR-73 TX (SSL) @ 2850 MHz vs. NEXRAD RX
2. NEXRAD TX vs. WDR-73 RX (NSSL) @ 2850 MHz

NEXRAD tuned above 2950 MHz - no interactions with an I/N > 9 dB were identified for both NEXRAD TX vs. environmental receivers and environmental transmitters vs. NEXRAD RX. Plots of I/N are provided for interactions with an I/N > -10 dB. (Enclosures D8-D9)

No RACONS were identified in the subject environment.

The following equipments in the environmental listing are to be replaced by ASR's as listed in TABLE 1.⁴ The frequency of the replacement equipment is not known but the newer equipment will include signal processing capable of interference rejection and, therefore, be less susceptible to NEXRAD interference.

TABLE 1
FAA ASR EQUIPMENT REPLACEMENT SCHEDULE

City	Present System	Replacement System	Target Date
Oklahoma City, OK (FAA Academy)	ASR-4	ASR-9	87/12
Oklahoma City, OK (Will Rogers Airport)	ASR-8	ASR-9	89/01
Tulsa, OK	ASR-5	ASR-9	87/05
Oklahoma City, OK (Depot)	None	ASR-7	87/12
Oklahoma City, OK (Depot)	None	ASR-7	90/01

V. CONCLUSIONS/RECOMMENDATIONS

To satisfy the requirements for NEXRAD to operate in the 2700-2900 MHz band, a primary frequency of 2890 MHz is recommended for the NEXRAD OSF. This recommendation is conditioned on coordination to preclude simultaneous operation of the NEXRAD OSF and the WDR-73 at the NSSL (usually referred to as the NRO Radar). Coordination is viable since neither the OSF nor the NRO Radar operate on a continuous basis. The NRO Radar was installed to perform research for the NEXRAD project and is only operated from April thru June during severe weather.

A secondary frequency of 2970 MHz or above is recommended for the OSF. This recommendation is based on an assigned frequency of 2950 MHz for the Oklahoma City NEXRAD which will be installed at nearby Twin Lakes.⁵ The 20 MHz frequency separation reduces the possibility of NEXRAD-to-NEXRAD interference.⁶

Signal processing was not considered for this analysis since final characteristics for NEXRAD have not yet been determined. Signal processing would be expected to reduce interference to NEXRAD.

IV. LIST OF ENCLOSURES

- A. Environmental listing of equipment in the 2700-3000 MHz band.
(Not included with this appendix.)
- B. One histogram of number of assignments vs. frequency (2700-3000 MHz band).
- C. One histogram of power per channel vs. frequency (2700-3000 MHz band).
- D. Nine plots of I/N vs. frequency.

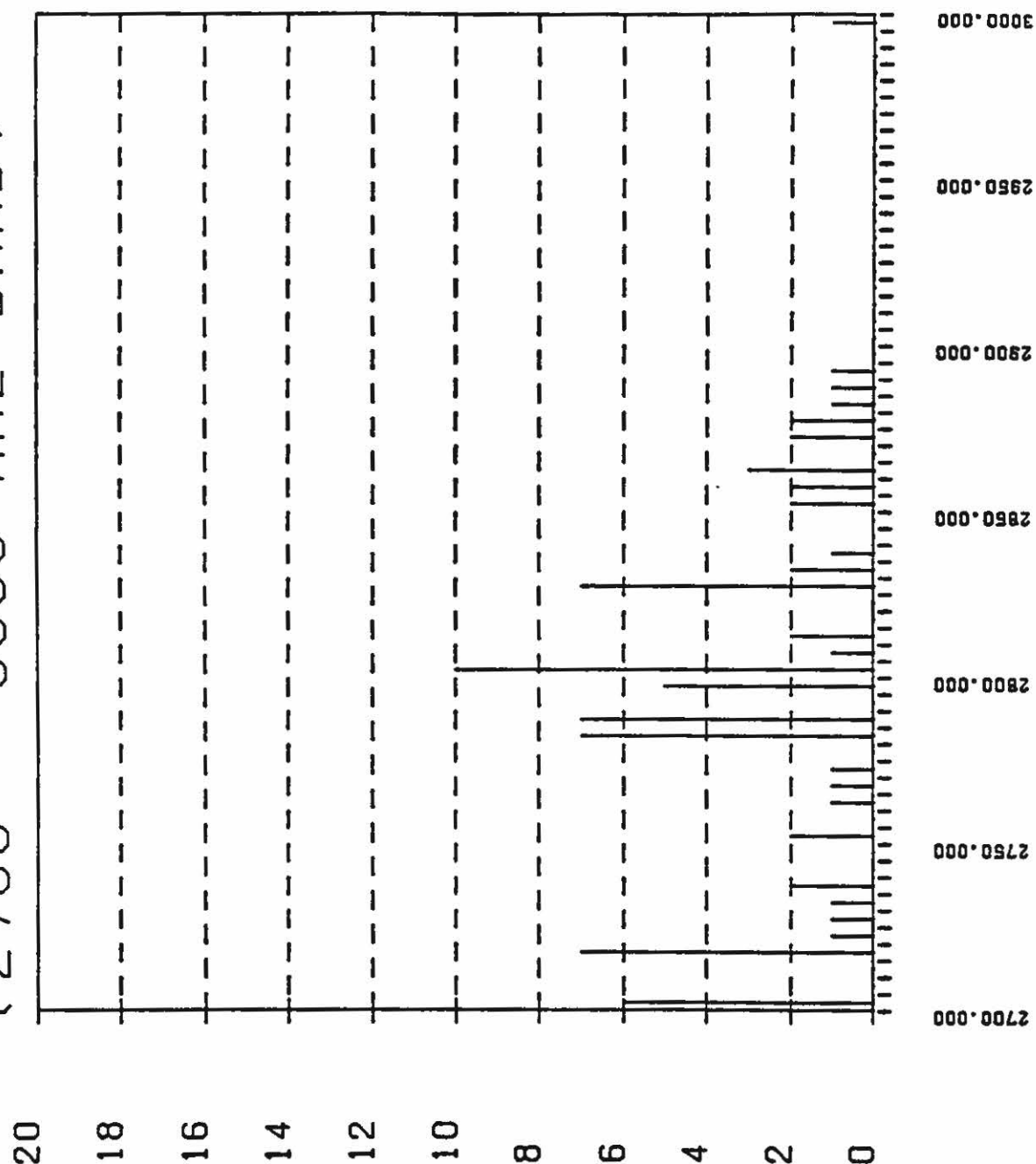
REFERENCES:

1. Hinkle, Robert L., Background Study on Efficient Use of the 2700-2900 MHz Band, NTIA Report 83-117, April 1983.
2. Letter from Juris Petriceks of SRI International, Menlo Park, CA dated 4 April 1986
3. Koppenhaver, et al., EMI Impact of Introducing NEXRAD in the 2900-3000 MHz Band, ECAC-CR-84-058, DoD ECAC, Annapolis, MD, July 1985.

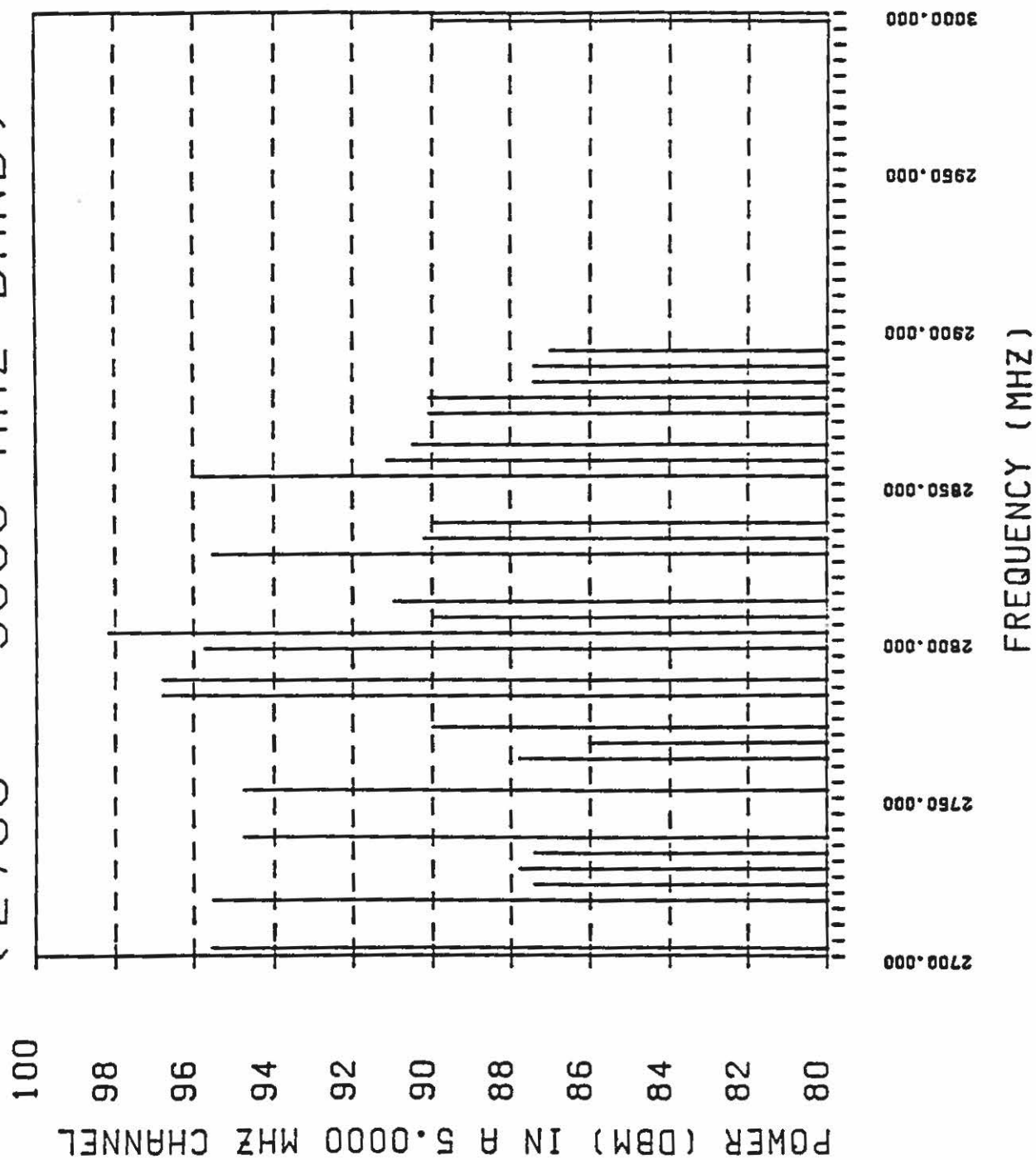
4. FAA list entitled ASR-9/Leapfrog Program Site Delivery Summary; revised 6/25/85 from Primeggia
5. Snyder, Larrey A., Oklahoma City NEXRAD Site Frequency Analysis, Annapolis, MD, May 1986.
6. Sangston, J. and Schwartz, D., Technical Characteristics and Frequency - Distance Relationships for NEXRAD and Other Systems in the 2700-3000 MHz Band, ECAC-CR-84-153, DoD ECAC, Annapolis, MD, February 1985.

NORMAN OSF ENVIRONMENT (2700 - 3000 MHZ BAND)

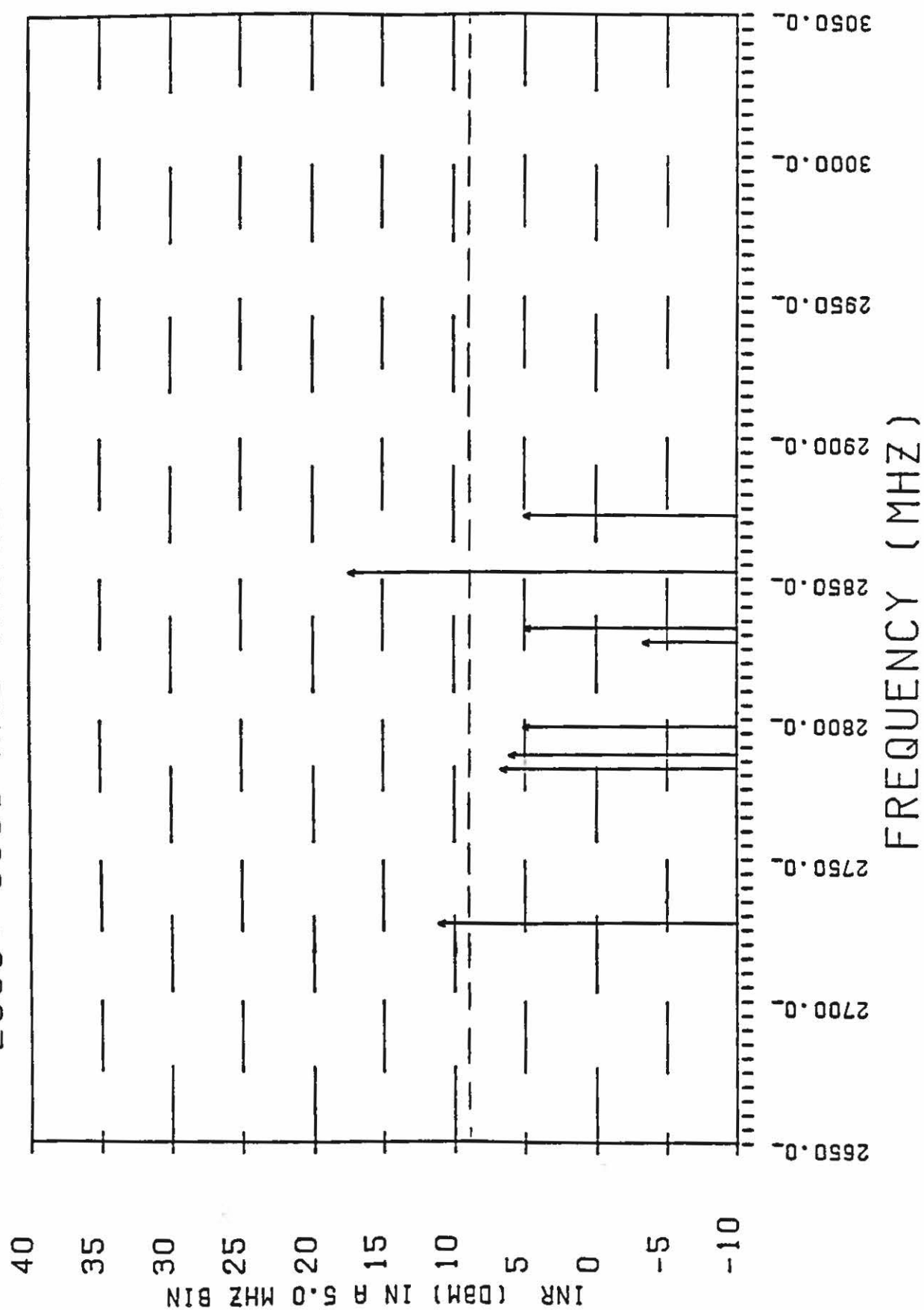
ASSIGNMENTS OCCUPYING A 5.0000 MHZ CHANNEL



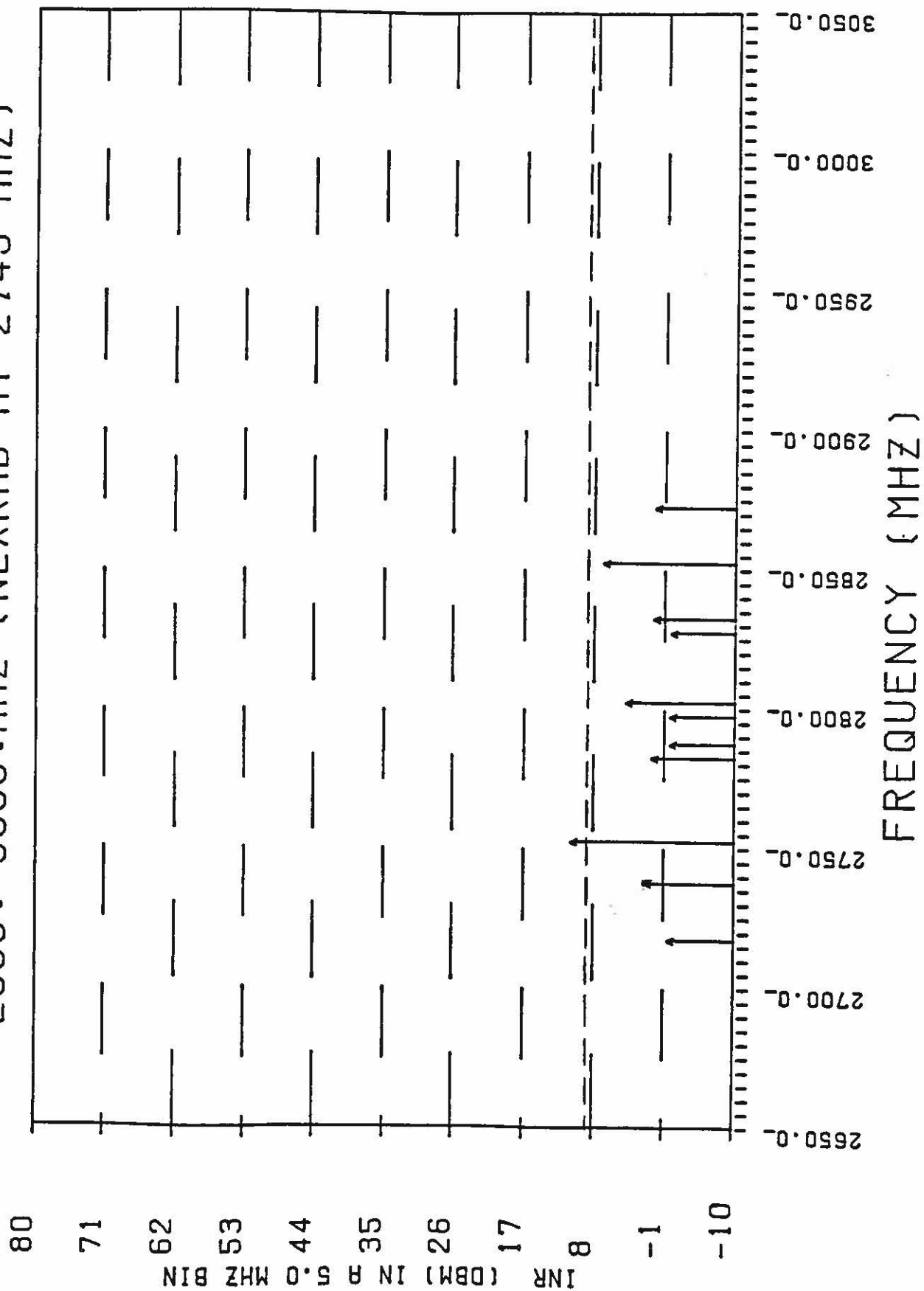
NORMAN OSF ENVIRONMENT (2700 - 3000 MHZ BAND)



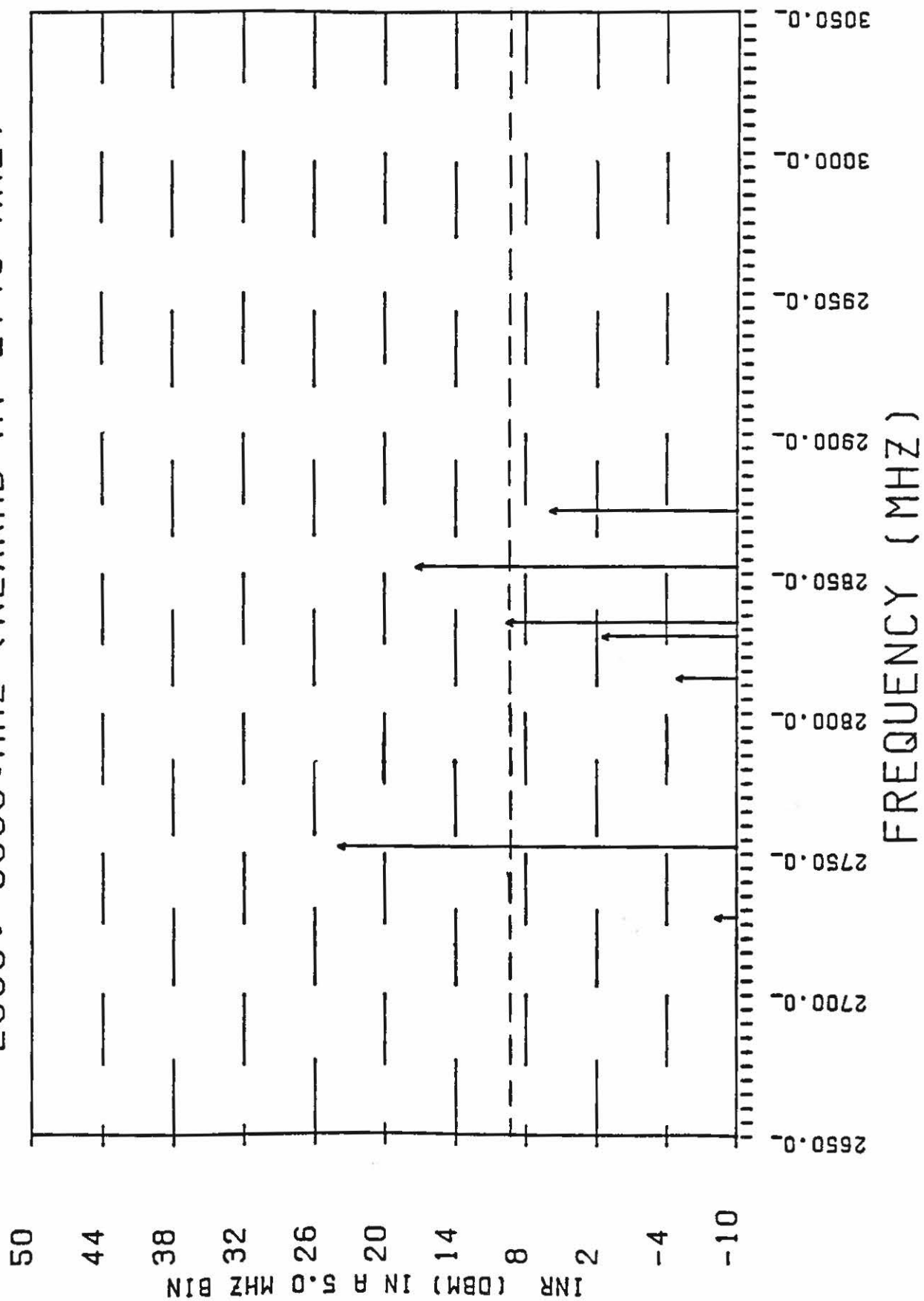
NORMAN TXS VS NEXRAD RX 2650.-3050.MHZ (NEXRAD AT 2705 MHZ)



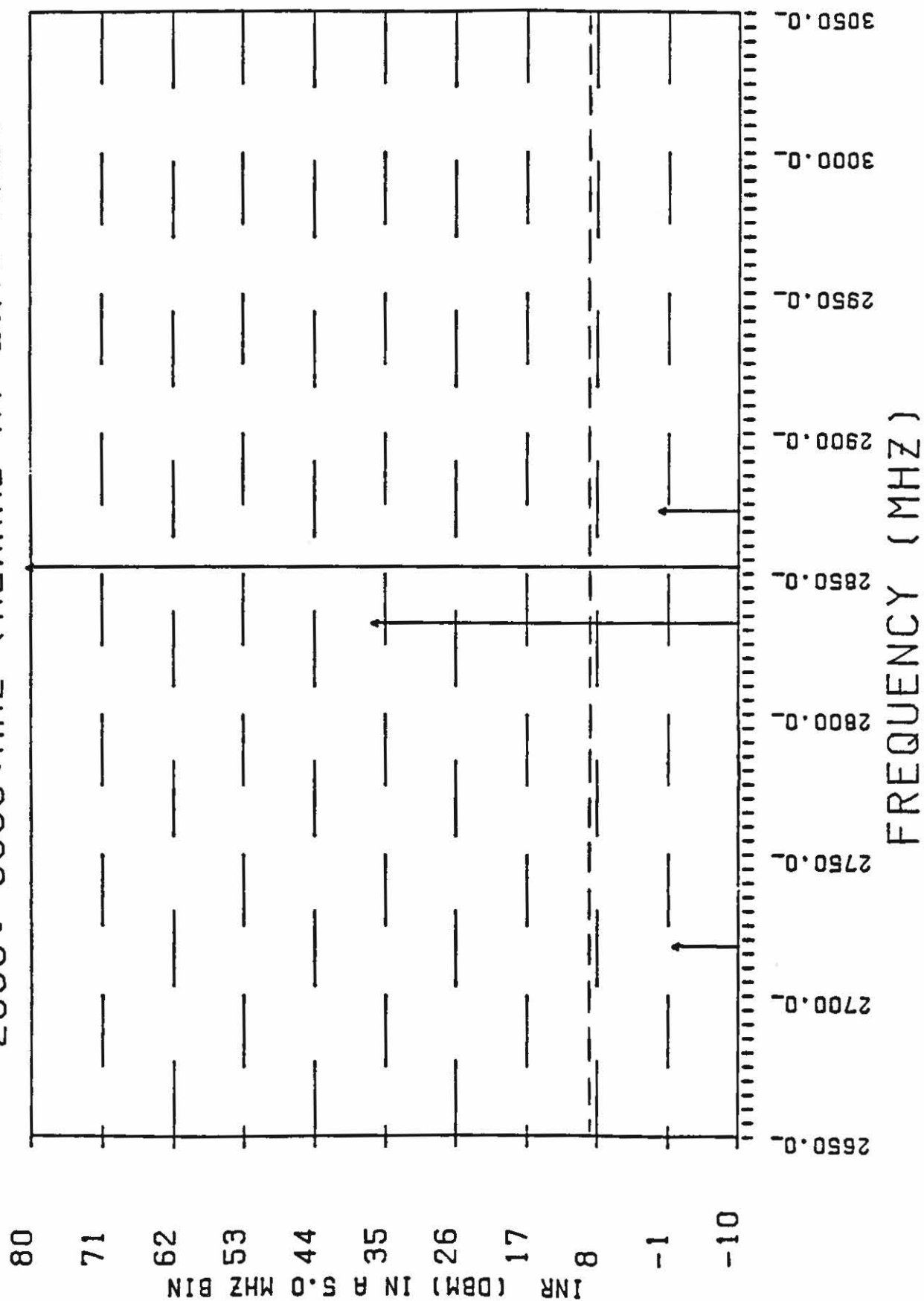
NEXRAD TX VS NORMAN RXS 2650.-3050.MHZ (NEXRAD AT 2745 MHZ)



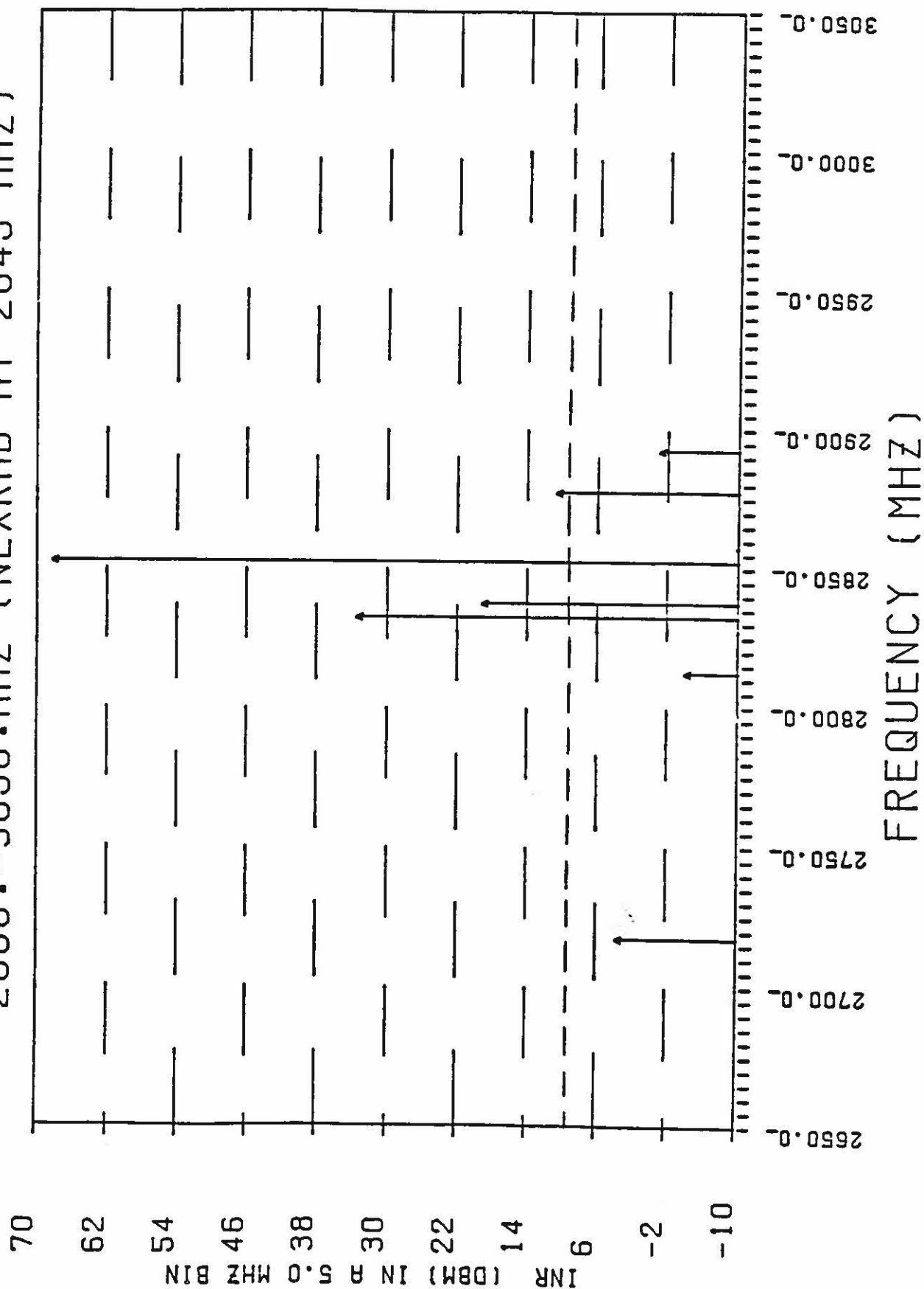
NORMAN TXS VS NEXRAD RX 2650.-3050.MHZ (NEXRAD AT 2745 MHZ)



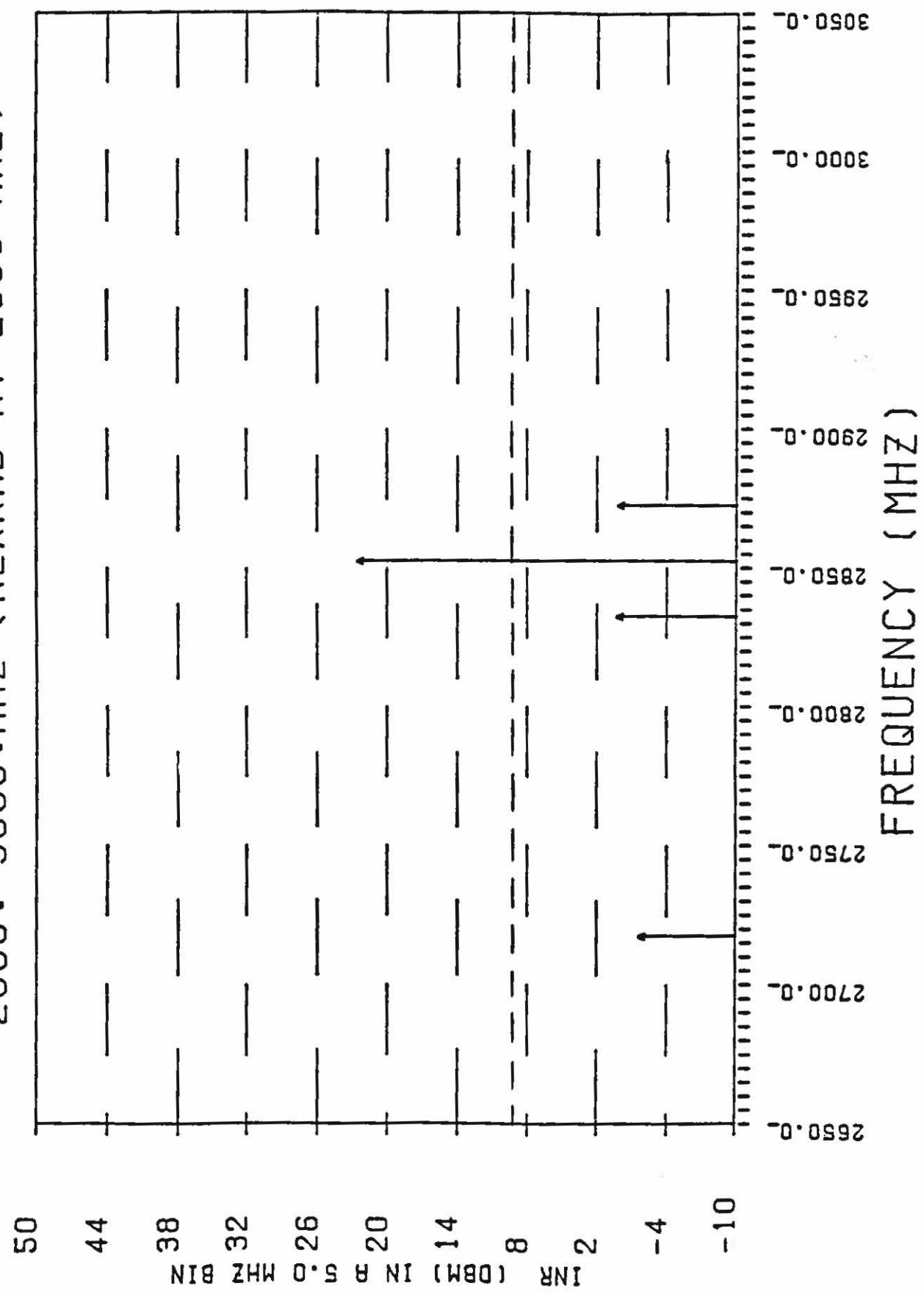
NEXRAD TX VS NORMAN RXS 2650.-3050.MHZ (NEXRAD AT 2845 MHZ)



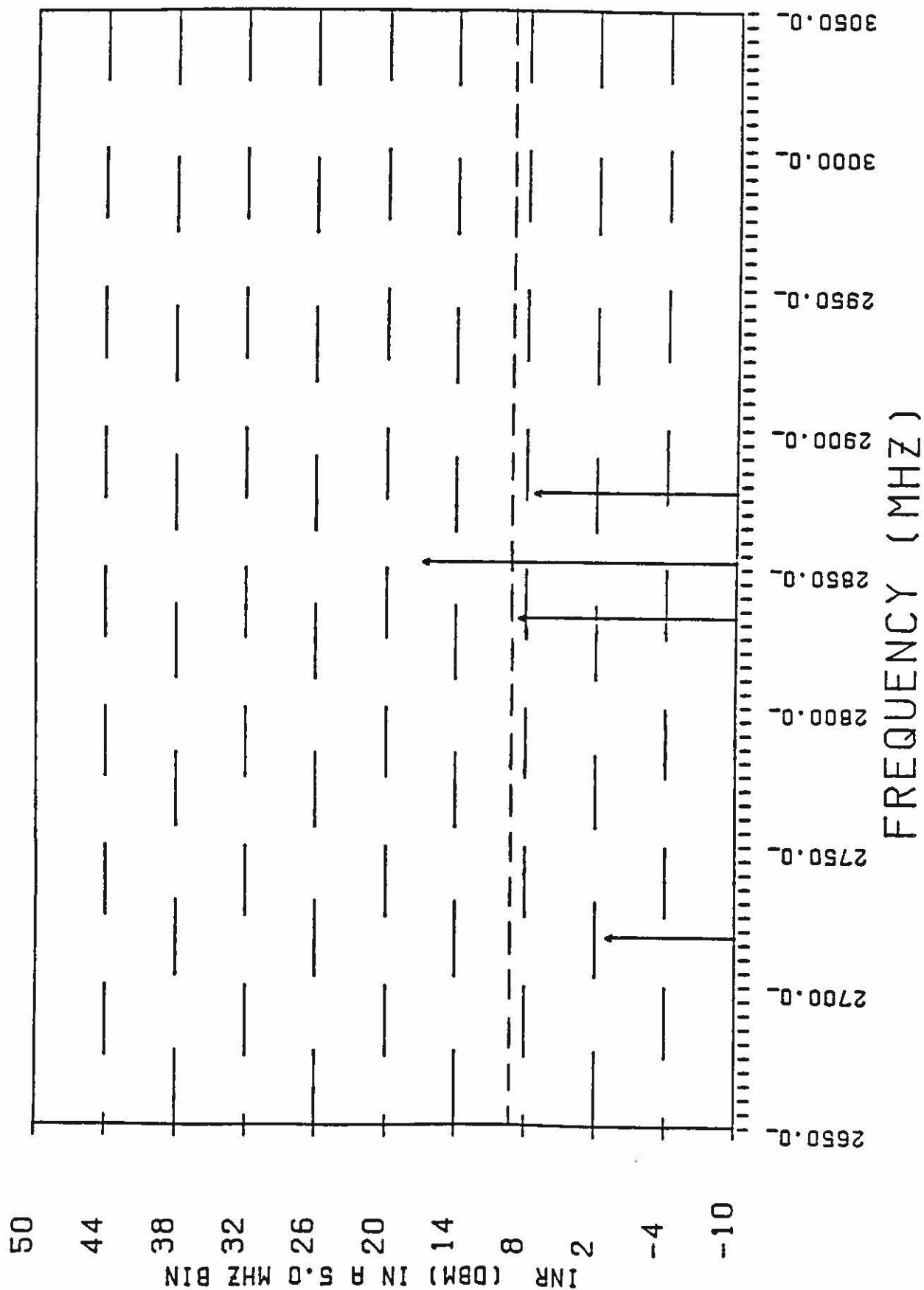
NORMAN TXS VS NEXRAD RX 2650.-3050.MHZ (NEXRAD AT 2845 MHZ)



NEXRAD TX VS NORMAN RXS 2650.--3050.MHZ (NEXRAD AT 2890 MHZ)



NORMAN TXS VS NEXRAD RX 2650.-3050.MHZ (NEXRAD AT 2890 MHZ)



NEXRAD TX VS NORMAN RXS 2650.-3050.MHZ (NEXRAD AT 2950 MHZ)

